



A meta-analysis of the relationship between milk protein production and absorbed amino acids and digested energy in dairy cattle

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ABSTRACT

Milk protein production is the largest draw on AA supplies for lactating dairy cattle. Prior NRC predictions of milk protein production have been absorbed protein (MP)-based and used a first-limiting nutrient concept to integrate the effects of energy and protein, which yielded poor accuracy and precision (root mean squared error [RMSE] >21%). Using a meta-data set gathered, various alternative equation forms considering MP, absorbed total EAA, absorbed individual EAA, and digested energy (DE) supplies as additive drivers of production were evaluated, and all were found to be superior in statistical performance to the first limitation approach (RMSE = 14%–15%). Inclusion of DE intake and a quadratic term for MP or absorbed EAA supplies were found to be necessary to achieve intercept estimates (nonproductive protein use) that were similar to the factorial estimates of the National Academies of Sciences, Engineering, and Medicine (2021). The partial linear slope for MP was found to be 0.409, which is consistent with the observed slope bias of –0.34 g/g when a slope of 0.67 was used for MP efficiency in a first-limiting nutrient system. Replacement of MP with the supplies of individual absorbed EAA expressed in grams per day and a common quadratic across the EAA resulted in unbiased predictions with improved statistical performance as compared with MP-based models. Based on Akaike's information criterion and biological consistency, the best equations included absorbed His, Ile, Lys, Met, Thr, the NEAA, and individual DE intakes from fatty acids, NDF, residual OM, and starch. Several also contained a term for absorbed Leu. These equations generally had RMSE of 14.3% and a concordance correlation of 0.76. Based on the common quadratic and individual linear terms, milk protein response plateaus were predicted at approximately 320 g/d

of absorbed His, Ile, and Lys; 395 g/d of absorbed Thr; 550 g/d of absorbed Met; and 70 g/d of absorbed Leu. Therefore, responses to each except Leu are almost linear throughout the normal in vivo range. De-aggregation of the quadratic term and parsing to individual absorbed EAA resulted in nonbiological estimates for several EAA indicating over-parameterization. Expression of the EAA as g/100 g total absorbed EAA or as ratios of DE intake and using linear and quadratic terms for each EAA resulted in similar statistical performance, but the solutions had identifiability problems and several nonbiological parameter estimates. The use of ratios also introduced nonlinearity in the independent variables which violates linear regression assumptions. Further screening of the global model using absorbed EAA expressed as grams per day with a common quadratic using an all-models approach, and exhaustive cross-evaluation indicated the parameter estimates for BW, all 4 DE terms, His, Ile, Lys, Met, and the common quadratic term were stable, whereas estimates for Leu and Thr were known with less certainty. Use of independent and additive terms and a quadratic expression in the equation results in variable efficiencies of conversion. The additivity also provides partial substitution among the nutrients. Both of these prevent establishment of fixed nutrient requirements in support of milk protein production.

Key words: model, amino acid, lactation, response

INTRODUCTION

Perhaps due to the historical pricing schemes, most dairy nutrition models have only provided predictions of milk volume (NRC, 1966, 1971, 1978, 1989, 2001). Thus, one must assume milk composition to assign an economic value to the predicted milk production, and changes in composition that may be induced by the new ration are not prospectively considered. This is perhaps a greater deficiency for milk fat production given relatively high variation in milk fat concentration than for milk protein concentration, but the latter does vary, and

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the variation is economically important. Additionally, the biological order of operation is reversed when predicting milk volume first because milk volume is a function of the osmotic pull of the milk components with most of the pull being generated by lactose (Jenness, 1974). Because lactose is synthesized from glucose by the enzyme lactose synthase which is composed of galactosyl-transferase and α -lactalbumin (Ebner and Schanbacher, 1974) with the latter being rate limiting (Jenness, 1974), lactose concentration and milk volume are driven by milk protein production which explains the low relative variation in milk protein concentration. Milk fat does not have significant osmotic potential, and thus it is much more variable in concentration. Predicting milk components first and using those to predict milk volume follows the biological process and is thus a more appropriate prediction approach. Additionally, that approach provides predictions of the primary value components of milk for use in nutrition models so that diets can be optimized to maximize product value minus input costs.

Several empirical and mechanistic models of milk protein production have been developed (Baldwin et al., 1994; Maas et al., 1997; Hanigan et al., 2001; Hristov et al., 2005a; Vyas and Erdman, 2009; Lean et al., 2018; Lapierre et al., 2020; Lapierre et al., 2023). Early efforts followed the conceptual scheme of Mitchell and Block (1946) where a single AA was assumed to be the sole limiter of milk or milk protein production. This approach was introduced for absorbed Lys and Met in NRC (2001). However, such an approach was shown to be an inappropriate representation at the tissue level (Hanigan et al., 2000). Over the past 20 yr, the role of regulation of ribosomal protein synthesis has been shown to be an important determinant of protein synthesis (Davis et al., 2002; Arriola Apelo et al., 2014b; Luo et al., 2018). One of the signaling pathways involved in regulation of protein synthesis, mTOR, responds independently and additively to several EAA, energy, and hormonal stimulation (Rius et al., 2010b; Appuhamy et al., 2012, 2014b; Arriola Apelo et al., 2014c; Castro et al., 2016b). Protein synthesis is also regulated by phosphorylated eIF2 α , which is mediated by general control nonderepressible2 kinase in response to EAA supplies (Anthony et al., 2004). Thus, the concept of a single limiting AA or nutrient is not supported by the observational data. Additional work has demonstrated that milk protein gene expression also changes in response to AA and energy supply (Toerien et al., 2010; Rius et al., 2010a; Dessauge et al., 2011).

Although it has been demonstrated that representation of the independent effects of each AA provides a better prediction of milk protein production at the tissue level (Hanigan et al., 2000), such an approach has not been widely adopted at the whole animal level (Lean et al.,

2018). Certainly, it represents a major conceptual shift from application of the first-limiting nutrient approach (Mitchell and Block, 1946) with AA supplies expressed as proportions of MP or total absorbed EAA. The latter may have contributed to the lack of progress as that representation mathematically creates non-linearity in the independent variables, interactions among AA, and variation in the responses to each at a given AA supply due to variation in the denominator. Adding or subtracting a single AA results in changes in the ratios of all other AA given the change in MP or EAA supply elicited by the addition or subtraction. Changes in the ratios of the nonmanipulated AA represent false signals as their supply was not altered by the single AA addition.

Following from the cell and tissue work, the objectives of this work were: (1) to assess the existing framework for prediction of milk protein production; (2) to develop and assess alternative, static, deterministic predictions of milk protein production using predicted absorbed nutrient supplies and observed animal characteristics as independent variables. It was hypothesized that the prediction model would be significantly improved when the supplies of digested energy (DE) and absorbed EAA were considered, and that expressing absorbed AA in absolute, additive terms would be superior to expressing relative to total absorbed EAA, MP, or DE.

MATERIALS AND METHODS

The work consisted of a mixed-model, meta-regression analysis of literature data. All data handling and statistical analyses was completed using R (v. 4.0.3; R Core Team, 2020) within R Studio (v 1.3.1093).

Meta Data

Data used for the analyses was assembled from datasets collected by the NRC (2001) committee with enhancements and additions by Bateman et al. (2008), Roman-Garcia et al. (2016), and Feng et al. (2018), the mammary data described by Hanigan et al. (2002), the total-tract fatty acid digestibility data of Daley et al. (2020), and data collected from studies describing the effects of infused AA and ruminally protected AA identified by a comprehensive literature search. In the latter case, the key words AA, dairy, cattle, and lactation were used to search within the Agricola and PubMed databases in 2019. Duplicates of studies already contained in the prior databases were excluded. Data were entered into an Excel-based database prepopulated with a single data format. In some cases, more than one experiment was described within a publication. In this case, each experiment was assigned a unique publication ID (**PubID**) assuming

the experiments were independent. The complete dataset contained 1153 treatment means from 273 experiments.

From the total observations, 933 treatments from 221 experiments reported milk protein production or it could be derived from data that were reported, and were thus selected. Three treatments examining the effect of infused insulin were dropped. Seven treatments testing the effects of DL-2-hydroxy-(4-methylthio)-butyrate and its isopropyl ester derivative were removed due to uncertainty regarding the bioavailability of those Met analogs.

Dietary nutrient inputs were derived from the reported dietary ingredients for each treatment using the feed library from NASEM (2021) with adjustment for reported ingredient and dietary nutrient composition as described by Hanigan et al. (2021) and Fleming et al. (2019b). Three studies containing 12 treatments were removed because dietary nutrient composition derived by summation of the ingredient nutrients were more than 3 SD from the reported dietary nutrient concentrations suggesting data reporting may have been inaccurate in the publication. After initial work, 1 experiment containing 4 treatments was removed, as the reported milk protein production was identified as an outlier based on variance in the residual errors generated from model predictions. Two additional treatments from another experiment were removed due to missing input values, yielding 905 treatment means from 216 experiments that were used for the analyses. A listing of the studies describing the experiments that were used is provided in Supplemental Table S1 (see Notes).

Seven hundred forty-six of the 905 treatments were from purebred Holstein, 7 from British Friesian, 4 from Jersey, 74 from Ayrshire, 4 from Brown Swiss, 8 from Polled Red or Red and White, and 4 from Pie Noir breeds. The remainder were Holstein crosses with Jersey, Simmental, or Brown Swiss breeds.

Milk protein was reported on a crude basis for 79 experiments, and the basis was unspecified for 76 experiments. In the latter case, 55 of the experimental descriptions indicated the analyses were conducted by infrared, generally at commercial laboratories. The milk analyzers in the US DHIA laboratories were calibrated to milk CP before the year 2000, and converted to a true protein (TP) basis thereafter (D. Barbano, Cornell University, Ithaca, NY, personal communication). Therefore, publications before 2000 were assumed to have been reported on a CP basis if unspecified, and thereafter on a TP basis. After characterization of the basis of measurement, all CP values were converted to a TP basis assuming TP was 95.1% of CP (DePeters and Cant, 1992).

Five experiments containing 16 treatments did not report milk fat concentration or production. These were filled with the mean milk fat concentration from the meta-data which was $3.60\% \pm 0.51$.

Parity was defined as a continuous value based on the reported animal composition of the treatments with nulliparous animals coded as 0 (unused for the work herein), primiparous animals coded as 1, and multiparous animals as 2. Twenty-nine experiments containing 110 treatment means did not report parity. For those treatments, 1 was entered for treatments with reported BW less than the mean BW for primiparous animals (525 kg), 2 for treatments with reported BW greater than the mean BW for multiparous animals (610 kg), and the mean group parity (1.842) for the remaining missing values.

Body weight was not reported for 43 experiments representing 157 treatment means. Missing BW for these treatments were predicted using a regression equation developed from observed BW and reported parity (before replacement of missing parity values):

$$\widehat{BW} = 552 \pm 29 + 59.5 \pm 8.7 \times (\text{Parity} - 1). \quad [1]$$

The residual SE was 53 kg; the adjusted R^2 was 0.083, and the slope probability was $P < 0.0001$. All observations contained values after application of Equation [1]. The mean BW before and after filling were 603 ± 58 and 602 ± 53 , respectively.

Twenty-eight studies encompassing 105 treatments did not report DIM. These were filled with the mean value of 138 ± 58 DIM.

The supplies of DE, MP, and absorbed EAA to the animal were predicted from diet composition and reported DMI using the nutrient supply equations of NASEM (2021). Briefly, ingredient fatty acids (FA) and the digestibility of FA were predicted from Daley et al. (2020); microbial protein, RDP, and RUP outflows from the rumen were from equations 52 and 26 of Hanigan et al. (2021), digested RUP and absorbed EAA were predicted as described by Fleming et al. (2019ab) with minor updates provided by NASEM (2021); ruminally digested fiber (RDNDF, % of DM) and starch (RDSt, % of DM) were calculated from the ruminal digestion coefficients described by White et al. (2016b); total-tract digested fiber and starch were calculated from de Souza et al. (2018); and total-tract digested residual OM from Tebbe et al. (2017). Digested energy intake (DEIn, Mcal/d) was calculated by summation of the apparently digested nutrients multiplied by enthalpies of 5.65, 9.4, 4.2, 4.0, and 0.89 Mcal/kg for digested protein, fatty acids, carbohydrate, residual OM (rOM), and urea, respectively (NASEM, 2021).

Because MP is included in DEIn, collinearity between DEIn and MP intake is introduced which is problematic for separating the effects of each. Similar problems are introduced for partitioning DEIn effects based on substrate type. Therefore, DEIn was parsed to

the individual contributions from starch, NDF, TP, FA, and rOM (**DEStIn**, **DENDFIn**, **DETPIn**, **DEFAIn**, and **DErOMIn**, respectively; Mcal/d), and the latter used in regressions. Combinations of these were created ad hoc except for nonprotein DEIn (**DEInp**, Mcal/d) used by NASEM (2021), which was precalculated for use in regressions.

NRC 2001 Evaluation

The NRC (2001) NE_L and MP allowable milk protein predictions and the first-limiting nutrient approach were assessed using the pertinent equations of that model and the current data.

Regression Techniques

Regression analyses were conducted using the `lmer` function of the `lme4` package (R Development Core Team, 2020). Mixed models were fitted to the data by maximum likelihood with experiment ID (PubID) denoted as a random effect. The SE reported in many of the older publications were incorrect due to the use of Proc GLM from SAS which did not model covariance (Littell et al., 1998). This primarily affected the SE estimates for repeated measures and random effects models, although even the early estimates from studies that used neither were incorrect, and in the absence of covariance information, cannot be corrected. Therefore, the data were weighted by the square root of the number of observations represented in each treatment mean, which generally captures differences in experimental power, but does not fully capture the power of repeated measure designs nor reflect differences in precision among laboratories (see discussion by Hanigan et al., 2021).

Model performance was summarized by Akaike information criterion (AIC). Because the number of observations was large and fixed in size, there was no need to correct AIC for sample size, and it was thus calculated as:

$$AIC = -2(\log \text{likelihood}) + 2N_{\text{parms}}, \quad [2]$$

where N_{parms} represented the number of parameters in the model, and serves as a penalty for more complicated models. Differences among nested equations were assessed statistically using the `anova` function of the `stats` package in R as described by Chambers and Hastie (1992).

The concordance correlation coefficient (CCC), root mean squared error (RMSE), and its partition were calculated from the residuals unadjusted for random study effects (provided by the `predict` function of `lme4`) as described by Lin (1989) and Bibby and Toutenberg (1977),

respectively. The unadjusted values provide an estimate of the expected performance of the models on novel data where random study effects cannot be estimated a priori, which is the case for all applications of the model outside of the current data. RMSE and CCC were also calculated for the adjusted data (generated by the `predict` function of `lme4` using the full mixed model solutions) where an estimate of the contribution of random effects could be calculated by difference.

A subset of the candidate models that were biologically consistent, plausible, and ranked high based on AIC and the RMSE analyses were further assessed for parameter stability across solves, potential collinearity problems, and parameter stability across subsets of the data. The latter was achieved using exhaustive Monte-Carlo based, cross-evaluations, which consisted of 500 iterations of fitting the model to random selections of 85% of the data, and evaluation of the resulting model using the remaining 15% of the data. This was carried out using the `caret` (v. 3.0–8) and `mlbench` (v. 2.1–1) packages of R. Data selection for the split between training and evaluation was completely random with no consideration of study. Means and SD of model parameters and model fit statistics were calculated from the results of the 500 cross-evaluation runs. Variation in parameter estimates across solutions less than 50% of the mean were denoted as stable and reliable, between 50% and 70% as moderately stable and reliable, and variance greater than 70% as very unreliable and not well supported by the data. These categories were supported by probabilities of the estimates differing from 0 where parameters with CV less than 50% generally had mean probabilities less than 0.05; CV between 50% and 70%, had mean probabilities generally ranging from 0.05 to 0.20; and CV greater than 70% having mean probabilities generally greater than 0.20.

Global Model Forms

Preliminary work to identify the best global model (those containing all of the terms considered for inclusion) forms was conducted using additive linear and quadratic terms for CP intake (**CPIn**), MP intake (**MPIn**), total absorbed EAA intake (**EAAIn**), DEIn and their interactions, and this was subsequently extended to consider deaggregated protein and energy terms. Based on the preliminary evaluations and the study objectives, 3 global mixed effects model equation forms were formulated using different methods of representing the effects of absorbed AA on milk protein synthesis. They are described in the following sections.

Although DIM was found to be generally significant in those equations, only 5 studies had variation in DIM among treatments within the study (Polan et al., 1991;

Schwab et al., 1992; Choung and Chamberlain, 1993; Volden, 1999; Samuelson et al., 2001); but, none of the studies were designed to test for stage of lactation effects and thus all were confounded by time and diet. Inclusion of DIM in the current models resulted in highly non-biological intercepts (maintenance requirement) suggesting it was confounded with study effects that could not be addressed in the absence of any within study comparisons. Year of publication was also found to be significant with approximately a 6 g/year gain in milk protein output, however, inclusion of the factor did not affect other parameters, and because no data could be included for the future, models retaining the term would result in predictions outside of the range of data from the present going forward, which is of concern. Thus, DIM, and year of publication were excluded a priori from the global models.

Absorbed EAA Expressed in Grams per Day. The first global equation form used AA inputs expressed as grams of absorbed AA/d and considered both linear and quadratic effects for each EAA.

$$\begin{aligned} \text{Milk Protein (g/d)} = & \text{Arg} + \text{His} + \text{Ile} + \text{Leu} + \text{Lys} + \text{Met} \\ & + \text{Phe} + \text{Thr} + \text{Trp} + \text{Val} + \text{Arg}^2 + \text{His}^2 + \text{Ile}^2 + \text{Leu}^2 \\ & + \text{Lys}^2 + \text{Met}^2 + \text{Phe}^2 + \text{Thr}^2 + \text{Trp}^2 + \text{Val}^2 + \text{NEAA} \\ & + \text{InfEAA}_{\text{Bld}} + \text{DEStIn} + \text{DErOMIn} + \text{DENDFIn} \\ & + \text{DEFAIn} + \text{BW} + (\text{Parity} - 1) + \text{PubID}, \end{aligned} \quad [3]$$

where the individual EAA terms represented the absorbed supplies of each (g/d); NEAA (g/d) the absorbed supply of NEAA (g/d); InfEAA_{InBld} the vascular supply of infused EAA (g/d); and DEStIn, DErOMIn, DENDFIn, and DEFAIn, the DE arising from digested starch, rOM, NDF, and FA, respectively (Mcal/d). BW (kg) was centered to the mean value, and Parity was a continuous value as described above. InfEAA_{InBld} was added to allow for potential differences in responses to AA absorbed from the intestine versus jugular or arterial infusions. There were 6 studies that used vascular infusions; all were delivered post-splanchnic. In this case, first-pass use by the gut and the liver during absorption is bypassed. This may result in a difference in the post-splanchnic supply of 5% to 10% (MacRae et al., 1997; Hanigan et al., 2004; Fleming et al., 2019a).

Following initial work that demonstrated that most responses to individual absorbed AA were linear, the individual squared AA terms were collapsed into a single summative quadratic term $\Sigma(\text{EAA}_a^2)$ (g²/d), which was the sum of the squared individual absorbed supplies of each EAA ($a = 1$ to 10) reflecting the potential of overall saturation of protein synthesis or its regulation by the

combination of EAA and assumes a common effect size among the EAA:

$$\begin{aligned} \text{Milk Protein (g/d)} = & \text{Arg} + \text{His} + \text{Ile} + \text{Leu} + \text{Lys} + \text{Met} \\ & + \text{Phe} + \text{Thr} + \text{Trp} + \text{Val} + \text{NEAA} + \text{InfEAA}_{\text{Bld}} \\ & + \sum_{a=1}^{10} (\text{EAA}_a^2) + \text{DEStIn} + \text{DErOMIn} + \text{DENDFIn} + \text{BW} \\ & + (\text{Parity} - 1) + \text{PubID}. \end{aligned} \quad [4]$$

Absorbed EAA Expressed as a Proportion of Total EAA. Historically, EAA requirements and supply have been expressed as a percentage of MP or of total EAA. This approach implicitly assumes the primary driver of production is MP or total EAA, and the EAA must have an ideal composition to achieve the full animal potential at that level of MP. The global model for that approach using total EAA as a denominator for calculation of percentages was

$$\begin{aligned} \text{Milk Protein (g/d)} = & \text{EAA} + \text{Arg}\% + \text{His}\% + \text{Ile}\% \\ & + \text{Leu}\% + \text{Lys}\% + \text{Met}\% + \text{Phe}\% + \text{Thr}\% + \text{Trp}\% \\ & + \text{Val}\% + \text{NEAA}\% + \text{Arg}\%^2 + \text{His}\%^2 + \text{Ile}\%^2 + \text{Leu}\%^2 \\ & + \text{Lys}\%^2 + \text{Met}\%^2 + \text{Phe}\%^2 + \text{Thr}\%^2 + \text{Trp}\%^2 + \text{Val}\%^2 \\ & + \text{NEAA}\% + \text{InfEAA}_{\text{Bld}}\% + \text{DEStIn} + \text{DErOMIn} \\ & + \text{DENDFIn} + \text{DEFAIn} + \text{BW} + (\text{Parity} - 1) + \text{PubID}, \end{aligned} \quad [5]$$

where the individual AA were expressed as grams absorbed/g of total absorbed EAA/d $\times 100$. Total EAA (g/d) expressed in grams absorbed/d was also included to ensure conservation of mass. The absence of such a term would allow predictions of milk protein production and associated individual EAA in excess of supplies of MP and individual absorbed EAA which violates conservation of mass principles. InfEAA_{Bld}%, expressed as a percentage of EAAIn, was included as above to account for potential differences between absorbed and systemically infused AA due to first pass splanchnic tissue use.

EAA Expressed as a Ratio to DEInp. Swine and poultry requirements for EAA are often expressed as a ratio to energy intake to ensure that energy and EAA are matched within the diet, although this suffers the same criticism as for expressing AA as a ratio to MP or total EAA. For this application, we expressed each as a ratio of DEInp (g/Mcal). As DEInp reflects a mass intake per day, the ratios of individual EAA to DEInp algebraically also reflect mass quantities of absorbed EAA for each negating the need to specify total EAA in the model to maintain conservation of mass:

$$\begin{aligned}
\text{Milk Protein (g/d)} = & \frac{\text{Arg}}{\text{DEInp}} + \frac{\text{His}}{\text{DEInp}} + \frac{\text{Ile}}{\text{DEInp}} + \frac{\text{Leu}}{\text{DEInp}} \\
& + \frac{\text{Lys}}{\text{DEInp}} + \frac{\text{Met}}{\text{DEInp}} + \frac{\text{Phe}}{\text{DEInp}} + \frac{\text{Thr}}{\text{DEInp}} + \frac{\text{Trp}}{\text{DEInp}} + \frac{\text{Val}}{\text{DEInp}} \\
& + \left(\frac{\text{Arg}}{\text{DEInp}} \right)^2 + \left(\frac{\text{His}}{\text{DEInp}} \right)^2 + \left(\frac{\text{Ile}}{\text{DEInp}} \right)^2 + \left(\frac{\text{Leu}}{\text{DEInp}} \right)^2 \\
& + \left(\frac{\text{Lys}}{\text{DEInp}} \right)^2 + \left(\frac{\text{Met}}{\text{DEInp}} \right)^2 + \left(\frac{\text{Phe}}{\text{DEInp}} \right)^2 + \left(\frac{\text{Thr}}{\text{DEInp}} \right)^2 \\
& + \left(\frac{\text{Trp}}{\text{DEInp}} \right)^2 + \left(\frac{\text{Val}}{\text{DEInp}} \right)^2 + \frac{\text{NEAA}}{\text{DEInp}} + \frac{\text{InfEAA}_{\text{Bld}}}{\text{DEInp}} + \text{DEStIn} \\
& + \text{DErOMIn} + \text{DENDFIn} + \text{DEFAIn} + \text{BW} + (\text{Parity} - 1) \\
& + \text{PubID}.
\end{aligned}$$

[6]

Model Selection

Initial model exploration was conducted by backward elimination using the step function of the lmerTest package (v. 3.1–2) of R. Terms were sequentially removed from the global models until all terms had *P*-values less than or equal to 0.05. Because backward elimination can result in biased models (Dunkler et al., 2014), final model selection was conducted using an all-models approach encoded in the dredge function of the MuMin package (v. 1.43.17) of R. With this method, all combinations of terms in each global model were assessed, and the models ranked based on statistical performance. This resulted in a maximum of 28 fixed-effect terms (10 EAA, 10 EAA², NEAA (g/d), InfEAA_{Bld}, 4 energy intakes, BW, and parity) equating to 536 million possible combinations to be evaluated and ranked. Because such a large matrix could not be held and manipulated in memory by R, even with 4 terabytes of memory, each of the global models were first evaluated to identify fixed terms that could be eliminated based on backward elimination so that the global models could be reduced to no more than 22 terms, which equates to 4.2 million combinations.

Final equations were primarily presented in tabular form. Tables were constructed using the tab_model function of the sjPlot package (v. 2.8.4) of R. Equation numbers are of the form (Table.Equation) to aid in finding equations within the manuscript. The number to the left of the decimal point denotes the table number and to the right the equation within the table. Equations numbered with whole numbers are presented in the body of the text.

RESULTS AND DISCUSSION

A summary of the meta data is provided in Table 1. In general, the data had a large range spanning milk produc-

tion from 10.2 to 53.8 kg/d with DMI ranging from 9.1 to 31.8 kg/d. There were 2 limitations to the data which were the general lack of characterization of pregnancy state and stage and the limited reporting of BW change. Studies often reported ending BW and compared differences among treatments, but the beginning BW were not generally reported. These deficiencies prevented meaningful assessments of the impact of BW change and gestational nutrient use on production and efficiency. The magnitude of that potential contribution to energy and protein supplies was examined by Liu et al. (2021).

All dietary nutrient intakes were normally distributed with a minimum range of 5 SD, but correlations among the inputs were generally high (>0.75; Supplemental Figure S1, see Notes) likely reflecting the scaling of all nutrients with milk production. Much of the range in EAA inputs was driven by trials which fed rumen-protected AA or infused AA (Supplemental Figures S2 and S3, see Notes). Although the infusion of proteins helps define the curvature at the top of the production range, the EAA within that infused protein are fixed in ratio and thus such treatments do not help address collinearity problems among individual EAA responses. Methionine absorption had the lowest correlations with other EAA ranging from 0.69 to 0.77 due to the large number of studies which explored the effects of methionine infusions and feeding ruminally protected methionine. The relatively large correlations among all of the other EAA underscores the value of infusion studies making use of single or small groups of EAA and feeding trials utilizing well characterized rumen-protected AA sources.

Digested energy intake and MPIn suffered from the same collinearity problem having a correlation coefficient of 0.83. Digested energy is composed of the energy contributed from each of the nutrients including MPIn, thus the 2 terms are mathematically related. Subtraction of the DE arising from MPIn from the DEIn term reduced the correlation between DEInp and MPIn to 0.69. The reduction was similar for the relationship between DEInp and total absorbed EAA, and between DEInp and each of the individual absorbed EAA (Supplemental Figure S1). The statistical benefits of removing the subnutrients of interest from the aggregated term was demonstrated for microbial protein flow predictions (Hanigan et al., 2021).

Absorbed supplies of each EAA were correlated with milk protein output, but the efficiency of conversion of MPIn to milk protein ranged widely and is very poorly related to the proportion of each EAA relative to total EAA (His, Lys, and Met relationships shown in Figure 1; the remainder not presented). MPIn conversion efficiency is also poorly related to each EAA expressed relative to DEIn (data not shown). Despite tremendous range in milk protein production per unit of MPIn or DEIn, there was no evidence of a break point for DEIn and only weak

Table 1. Descriptive statistics of the dataset used to derive model parameters and test the models (n = 905)

Variable ¹	Mean	Median	SD	Minimum	Maximum
Publication year	2000	2000	8	1974	2020
Observations/treatment	9.27	6.00	9.5	2	58
Parity (1 = primiparous, 2 = multiparous)	1.87	2.00	0.27	1	2
DIM	136	137	53	36	344
BW, kg	612	609	56	445	788
Milk production, kg/d	30.8	30.9	7.5	10.2	53.8
Milk true protein, %	3.02	3.02	0.21	2.45	3.71
Milk fat, %	3.59	3.58	0.50	2.11	6.60
DMI, kg/d	20.7	20.8	3.6	9.1	31.8
Dietary starch, % of DM	27.2	28.3	7.9	1.1	53.0
Dietary NDF, % of DM	32.8	32.2	5.9	14.6	54.3
Dietary forage NDF, % of DM	24.3	23.6	6.3	0.0	54.3
Dietary ADF, % of DM	19.9	19.4	3.9	7.2	35.8
Dietary lignin, % of DM	3.55	3.42	0.8	1.6	7.3
Dietary CP, g/d	3,459	3,467	809	1,075	5,944
Infused CP _{systemics} g/d	1.14	0.0	16.7	0	369
MP, g/d	1,981	2,000	393	751	3,271
Dietary CP, % of DM	16.4	16.6	2.3	8.3	25.8
A fraction, % of CP	38.1	36.8	8.0	21.2	81.5
B fraction, % of CP	52.9	54.0	8.6	13.2	73.5
C fraction, % of CP	8.93	8.44	2.6	4.2	18.1
Dietary fatty acids, % of DM	3.38	2.89	1.4	1.1	9.2
Dietary ash, % of DM	7.32	7.22	1.5	3.2	14.0
Dietary residual OM, % of DM	13.3	13.1	4.8	0.0	41.8
Digested starch, % of DM	24.0	24.8	6.9	1.0	48.0
Digested NDF, % of DM	17.0	16.3	4.7	5.5	35.9
Digested fatty acids, % of DM	2.21	2.05	1.1	0.05	6.42
Digested residual OM, % of DM	12.8	12.6	4.7	-2.9	40.1
Dietary RDP, % of DM	11.4	11.4	1.8	5.5	19.5
Dietary RUP, % of DM	5.08	5.00	1.2	1.9	11.5
Dietary MP, % of DM	9.6	9.5	1.1	5.8	15.4
DE, Mcal/kg	2.99	3.00	0.13	2.60	3.33
DEIn, Mcal/d	62.1	62.3	11	27.8	99.7
DEInp, Mcal/d	48.0	48.1	8.6	22.0	76.0
DEItp, Mcal/d	14.0	14.2	4.0	3.6	26.8
Fatty acid composition, % of DM					
C12:0	0.056	0.047	0.05	0.00	1.03
C14:0	0.040	0.034	0.03	0.01	0.23
C16:0	0.664	0.469	0.41	0.19	3.18
C16:1	0.033	0.018	0.03	0.00	0.26
C18:0	0.159	0.075	0.19	0.02	2.18
C18:1c	0.745	0.566	0.49	0.03	3.14
C18:1t	0.021	0.006	0.034	0.0001	0.26
C18:2	1.25	1.19	0.51	0.21	3.02
C18:3	0.33	0.29	0.16	0.10	1.71
Dietary AA intake, g/d					
Arg	165	164	55	32	333
His	83	85	25	23	169
Ile	145	147	41	40	271
Leu	287	295	85	94	575
Lys	159	163	54	31	306
Met	56	56	16	20	113
Phe	166	168	45	46	310
Thr	133	136	37	40	238
Trp	41	42	12	9	81
Val	180	183	48	54	349
Absorbed AA, g/d					
Arg	106	105	24	35	180
His	48	48	12	17	93
Ile	116	117	22	44	218
Leu	184	186	43	73	360
Lys	151	153	30	53	245
Met	46	47	10	18	82
Phe	115	116	24	42	186
Thr	105	106	20	40	165
Trp	26	26	5.3	9	43
Val	124	125	24	47	228
EAA	1,021	1,031	201	377	1,740
NEAA	1,258	1,268	256	487	2,022

¹DEIn = digested energy intake; DEInp = DEIn minus DEItp; DEItp = DEIn from digested true protein.

evidence for MPIn (Figure 1). Such points have classically been used to define a requirement. The apparent quadratic responses to Lys and Met were due entirely to a small number of rumen-protected AA feeding trials and infusions, and the range in milk protein within the mass of observations in the very middle of the input range was much greater than the predicted range based on the regression equations (Figure 1). The lack of a clear relationship was particularly apparent when expressing EAA supplies as a ratio with DEIn. Thus, the data offer little support for modeling milk protein responses as a function of the proportion of individual EAA in total EAA or as a ratio to DEIn.

NRC 2001 Model Performance

From Occam's Razor, models should generally be no more complicated than necessary, and thus an evaluation of the NRC (2001) model and its extension based on the first-limiting nutrient concept (Mitchell and Block, 1946) was undertaken as a reference point. Because the model does not output predictions of milk protein, these were generated by reversing the calculation of MP allowable milk using the observed milk protein concentrations for each observation (Table 2 and Figure 2). The resulting predictions show clear systematic bias as previously reported (White et al., 2016b) with the model generating predictions generally centered in the data for NE_L and MP allowable, but underpredicting by 91 g/d when considering the minimum of the 2 and by 202 g/d when considering the minimum of NE_L , MP, and all 10 EAA. All of the predictions exhibited negative slope biases ranging from -0.44 to -0.24 g/g indicating the model greatly overpredicted responses to changes in the supply of those nutrients. Although the slope bias marginally improved as more nutrients were considered in the first limiting nutrient framework, the increase in mean bias overwhelmed those gains resulting in predictions with much poorer accuracy and precision as measured by RMSE and CCC. Thus, the model has structural or parameter problems that preclude unbiased estimates of milk and milk protein production, and the problems do not derive from the nutrient supply equations as the feed library and the nutrient supply predictions used herein were demonstrated to be largely free of mean and slope bias (White et al., 2016a, 2017; Fleming et al., 2019a,b; Daley et al., 2020; Hanigan et al., 2021). Hanigan et al. (2000) observed the same mean bias problem when the first limiting nutrient concept was applied at the mammary tissue level, suggesting the concept lacks validity.

The slope bias associated with MP allowable protein indicated the marginal response of milk protein to MP input was approximately 40% rather than the 67% used

in the model. Because slope bias was reduced, but not eliminated, when the EAA were added to the calculation of first-limiting production, a portion of the problem may be due to the effect of dietary AA on milk protein production. However, the simultaneous introduction of mean bias as the concept is extended across nutrients clearly indicates that the form of expression is not reflective of the underlying biology. Adding a representation of a mechanism underpinning the aggregated function should never increase the error.

The poor performance of the NRC (2001) model should not be surprising. It has been clearly shown that marginal recovery of MPIn decreases as MPIn increases (Doepel et al., 2004; Metcalf et al., 2008). Therefore, usage of a fixed efficiency of utilization of MP or EAA directly leads to this type of slope bias (Lapierre et al., 2020). In addition, evidence began accumulating that milk protein production responded independently to multiple EAA simultaneously (Schwab et al., 1976; Clark et al., 1978). Hanigan et al. (2000) demonstrated the abject failure of the single-limiting EAA approach for predicting milk protein output from EAA removed by the mammary glands, and the clear superiority of representing the effects of each EAA in an additive form. This concept was reiterated by Cant et al. (2003), and a plethora of in vivo and in vitro cell signaling work has demonstrated that milk protein synthesis is highly regulated by at least insulin and IGF-1 concentrations, energy supply, and Ile, Leu, Lys, Met, and Thr with strong trends for several additional EAA (Burgos and Cant, 2010; Burgos et al., 2010; Rius et al., 2010c; Appuhamy et al., 2011b, 2012, 2014a; Arriola Apelo et al., 2014a; Liu et al., 2017; Yoder et al., 2020). However, the regulation has not always been observed (Cant et al., 2001; Doelman et al., 2015a,b; Nichols et al., 2017).

Protein and Total EAA Based Predictions

Given the evidence that the current framework is inadequate, work was undertaken to understand the general milk protein response surface with respect to MP and DEIn. Given the complexity of considering the effects of 10 or more individual AA responses, understanding the general response surface was deemed important.

Although CP-based models (NRC, 1971, 1978) are considered inferior to absorbed protein (NRC, 1989), and MP-based models (NRC, 2001), the improvement in precision across that spectrum is modest (Table 3, see Equations [3.1] to [3.8]). Equation [3.1] containing linear and quadratic terms for CPIn and a linear DEIn term performed nearly as well, based on AIC, as Equation [3.4] using linear and quadratic MPIn terms, the latter having a gain in precision of 22 g/d. The total EAA based equation performed the same as the MP-based equation

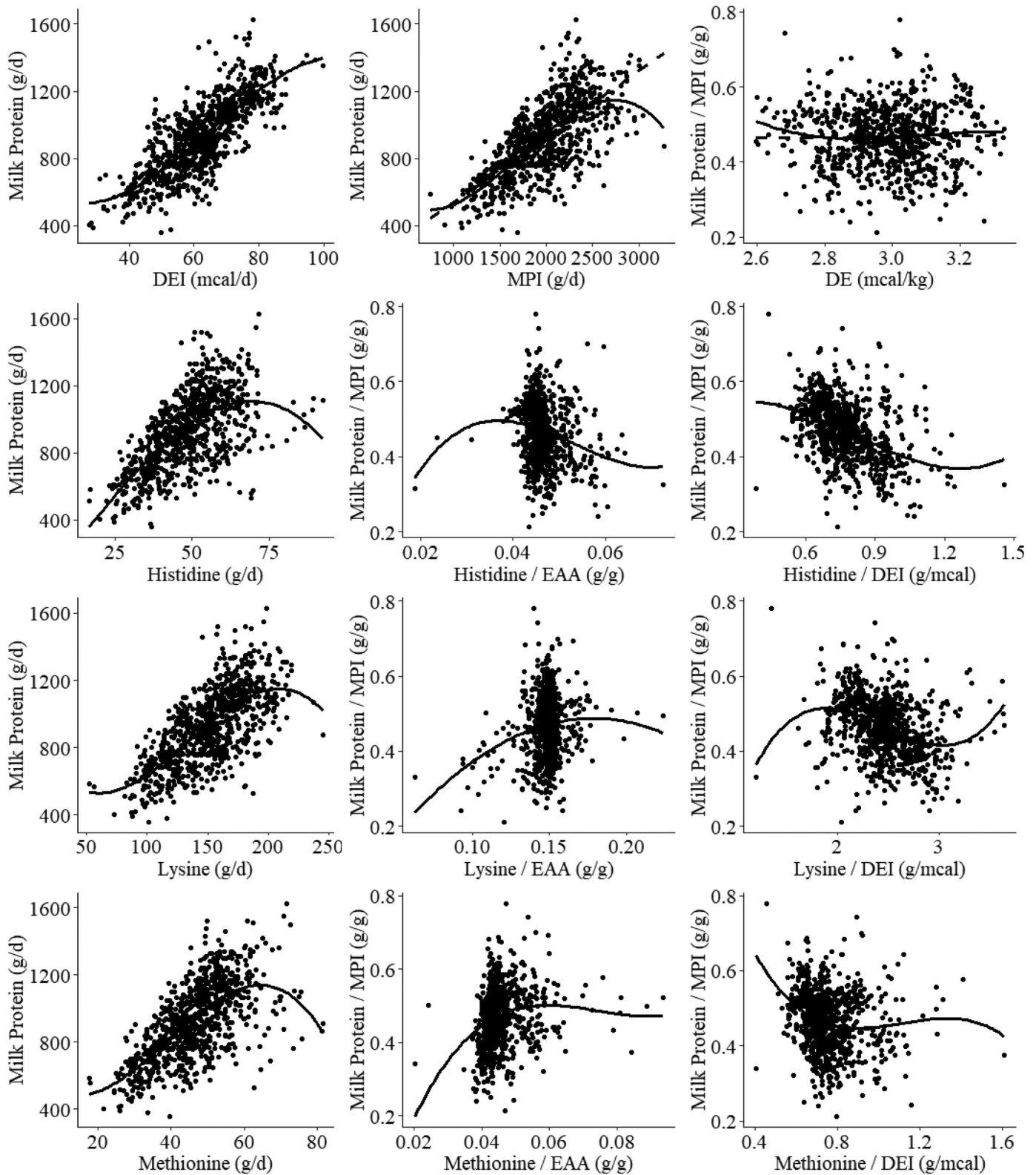


Figure 1. Observed milk true protein responses to digested energy (DE), DE intake (DEI), MP intake (MPI), absorbed His, absorbed Met, and absorbed Lys. EAA represented total absorbed EAA.

Table 2. Residual error analyses of NRC (2001) predictions of NE_L allowable, MP allowable, the minimum of NE_L and MP allowable, and the minimum of NE_L, MP, and each of the 10 EAA allowable predictions¹

Item ²	NE _L	MP	Minimum		
			NE _L and MP	EAA	All
N	905	905	905	905	905
Observed mean, g/d	922	918	922	918	922
Predicted mean, g/d	953	890	831	738	719
RMSE, g/d	194	228	210	264	269
RMSE, %	21.0	24.9	22.8	28.7	29.2
Mean bias, % MSE	3	2	19	47	57
Slope bias, % MSE	38	32	21	5	4
Mean bias, g/d	-31	28	91	180	202
Slope bias, g/g	-0.38	-0.44	-0.34	-0.26	-0.24
<i>P</i> _{Mean bias}	0.0001	0.0002	0.0001	0.0001	0.0001
<i>P</i> _{Slope bias}	0.0001	0.0001	0.0001	0.0001	0.0001
CCC	0.77	0.65	0.71	0.52	0.54

¹The EAA allowable predictions were calculated using the same framework as for MP but substituting each of the individual EAA to calculate supply, requirements, and thus allowable milk protein production.

²CCC = concordance correlation coefficient; RMSE = root mean squared error (g/d or % of observed mean); N = number of treatment means.

reflecting the very high correlation between MPIn and EAA intakes (Supplemental Figure S1).

Digestible energy intake was a very strong linear, additive determinant of milk protein production regardless of whether the DE from MP was included in the energy term (DEIn; Equation [3.4]) or excluded to reduce collinearity (DEInp; Equation [3.5]). Although the latter resulted in a very slight decrease in statistical performance, removal of the collinearity problem likely revealed the true independent effects of DEInp and MPIn. Considering that 22% of DEIn derives from MPIn, an estimate of 10.99 g protein/Mcal of DEIn (Equation [3.4]) equates to a partial slope for MPIn of 0.428 g protein/g MPIn ($10.99 \times 0.22/5.65$). Adding the observed linear slope coefficient for MPIn to the partial energy slope results in an overall response to changes in MPIn of 0.755 g protein/g MPIn ($0.428 + 0.327$) as compared with the derived slope of 0.409 g milk protein/g MPIn from Equation [3.5]. A slope estimate of 0.755 g/g is more than that used by NRC (2001), which resulted in a very large slope bias. Therefore, one can conclude the slopes from Equation [3.4] are nonbiological, presumably due to collinearity problems.

There was no evidence of an interaction between DEInp and protein terms (discussed further below). At a mean DEInp of 48.0 Mcal/d, the DEInp term in Equations [3.5] and [3.8] represented 530 and 532 g/d of milk protein, respectively, versus 794 and 812 g/d, respectively, associated with the CPIn and MPIn terms. Thus, if representative, approximately 40% of the driving force for milk protein production is from the nonprotein energy supply.

Stimulating milk protein output in the absence of a change in the EAA supply necessitates an improvement

in post-absorptive EAA efficiency. The meta-analyses of Hanigan et al. (1998, 2002) and experimental work of Bequette et al. (2000) demonstrated that mammary EAA transport activity responds reciprocally to supply, increasing activity when supply is low, and the reverse. Subsequent work by Fleming et al. (2019a) demonstrated that the liver acts in the opposite manner, increasing transport activity when arterial EAA concentrations are high, and the reverse when low. Acting in concert, the 2 tissues can shift more of the EAA supply to mammary use when milk protein production is stimulated by a greater supply of energy providing the needed flexibility in net supply to mammary tissue as DEIn changes (Vanhatalo et al., 2003; Raggio et al., 2006; Rius et al., 2010a).

Although the overall relationship between MPIn and milk protein production appears generally linear ($P < 0.01$, Figure 1), there were significant nonlinear relationships for all 3 approaches (CP, MP, and EAA; Table 3). This was not due solely to high leverage points at the upper range of the data, as the nonlinear terms cross-evaluated well with repeated, random removal of 15% of the data (discussed below). Representing the relationship between milk protein production and MPIn as a linear relationship resulted in significantly poorer prediction performance (Equation [3.2] vs. Equation [3.3]), and perhaps more importantly, a non-biological intercept (419 ± 28 g/d).

The negative value of the intercept should reflect nonproductive net protein (NP) requirements of a fasted animal, and thus the intercept would be expected to be significantly negative for all animals. Inclusion of both the linear and squared terms for MPIn (Equation [3.3]) yielded an intercept that did not differ from 0. When DEIn was also included in the regression, the intercept

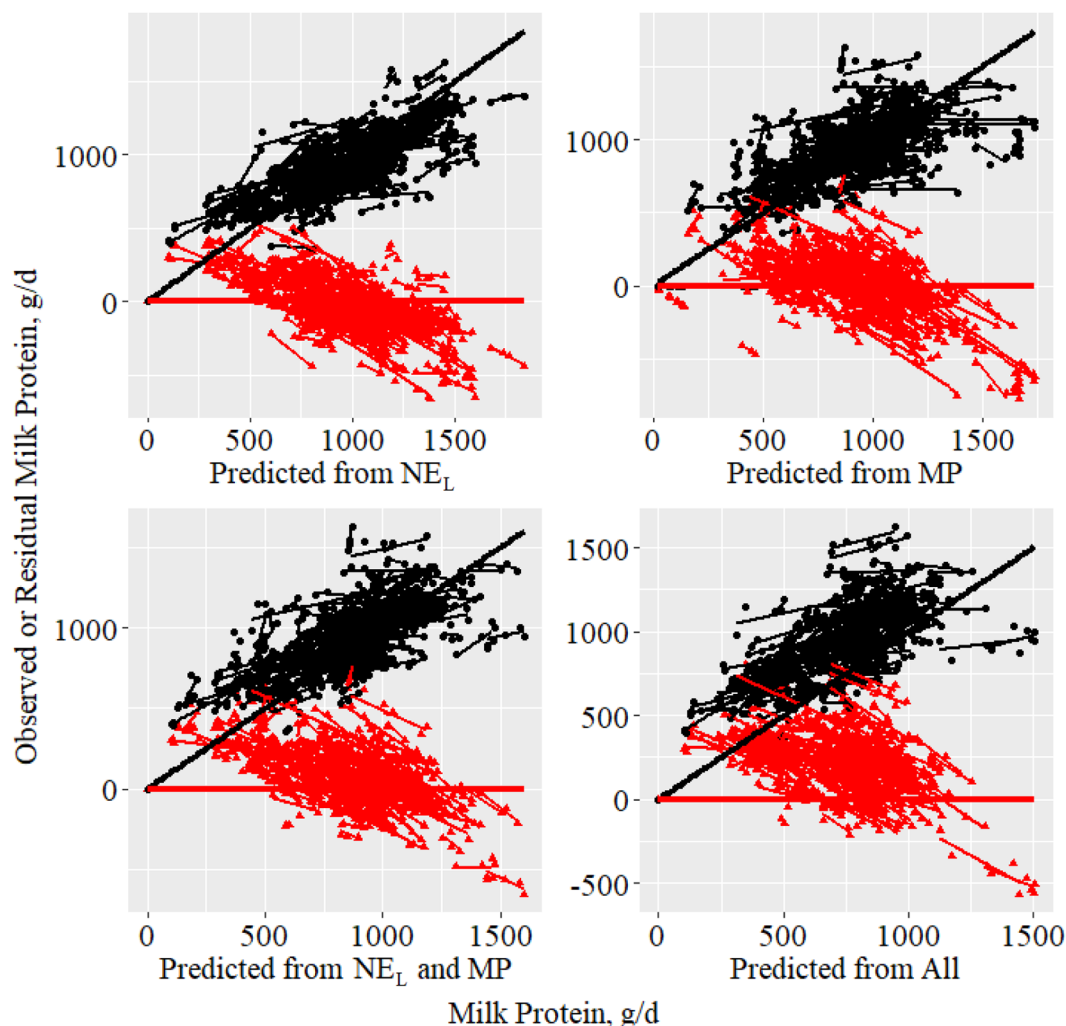


Figure 2. Observed (black) and residual errors (red) for milk protein production versus predicted based on NRC (2001) predictions of NE_L allowable, MP allowable, the minimum of NE_L and MP allowable, and the minimum allowable predictions from NE_L , MP, and each of the 10 EAA. The EAA allowable predictions were calculated using the same framework as for MP but substituting each of the individual EAA to calculate supply, requirements, and thus allowable milk protein production.

assumed a value of -150 g/d (Equation [3.4]). Use of DEInp to remove the effects of MP from the energy term yielded an intercept of -165 g/d NP/d (Equation [3.5]). This was less than estimates of 223 g/d predicted by NASEM (2021) for scurf and endogenous urinary NP losses by a fasted, nonproductive, 650 kg animal, but the difference was less than the SE of the intercept estimate.

From Equation [3.3], the efficiency of MP conversion to NP in a fasted state should be approximately 0.65 , which is consistent with NRC (2001) estimates. Because the slope presents the fractional capture of MP in milk NP, 1 minus the slope reflects the variable cost of production including metabolic fecal protein and inefficiencies of protein capture resulting in protein catabolism. At an efficiency of 0.65 , the MP intake required to meet endogenous urinary and scurf protein use would be 254

g/d. The difference of 89 g/d is less than the 127 g NP/d requirement for metabolic fecal protein predicted by NASEM (2021) for an animal at maintenance intake levels, but as DMI increases with increasing milk production, the estimated nonproductive NP use does align with the expected losses based on the inefficiency of conversion of MP to milk NP. Thus, the factorial and regression approaches do not perfectly align with respect to nonproductive protein requirements at maintenance intakes and likely at very low levels of production. Possible reasons for the misalignment include extrapolation errors given a minimum milk protein of approximately 300 g/d in the dataset, and use of a milk protein efficiency for estimation of nonproductive functions. If the independent effects of DEInp and MPIn in Equation [3.5] are accepted, the efficiency of conversion of MP to NP increases as

Table 3. Models relating MP, total EAA, and digestible energy to milk true protein production (g/d; n = 905)

Predictor ¹	Base [3.1]			Equation [3.2]			Equation [3.3]			Equation [3.4]			Equation [3.5]			Equation [3.6]			Equation [3.7]			Equation [3.8]		
	Estimate	SE		Estimate	SE		Estimate	SE		Estimate	SE		Estimate	SE		Estimates	SE		Estimate	SE		Estimate	SE	
Intercept	-190	52		419	28		47	74		-150	66		-165	67		-178	67		-130	65		-126	66	
BW - 612, kg																								
CPIn, g/d	0.234	0.031																						
MPIn, g/d				0.252	0.013		0.646	0.074		0.327	0.068		0.409	0.068		0.406	0.068		0.60	0.13		0.71	0.13	
EAA, g/d																								
CPIn ² (g/d) ²	-2.7 × 10 ⁻⁵	4.0 × 10 ⁻⁶																						
MPIn ² (g/d) ²																								
EAA ² (g/d) ²																								
DEIn, Mcal/d	10.38	0.74																						
DEInp, Mcal/d																								
AIC	10.411			10.707			10.680			10.430			11.04			11.43			10.425			11.08		
σ ²	6.941			9.362			9.027			7.154			7.414			7.370			7.102			7.349		
τ ₀₀ (PubID)	17,144			26,709			26,213			17,015			17,712			17,538			17,034			17,791		
ICC	0.71			0.74			0.74			0.7			0.7			0.7			0.71			0.71		
Marginal R ²	0.492			0.214			0.223			0.493			0.469			0.460			0.493			0.468		
Conditional R ²	0.854			0.796			0.801			0.850			0.843			0.840			0.851			0.845		
Obs mean, g/d	922			922			922			922			922			922			922			922		
Pred mean, g/d	917			919			919			924			925			927			924			925		
CCC	0.61			0.524			0.537			0.751			0.733			0.726			0.751			0.731		
RMSE, g/d	156			166			164			134			137			138			134			138		
RMSE, %	16.9			18.0			17.8			14.5			14.9			14.9			14.5			14.9		
Mean bias, %	0.1			0.0			0.0			0.0			0.0			0.1			0.0			0.0		
Slope bias, %	5.9			10.7			11.4			2.8			3.2			4.1			2.8			3.1		
Mean bias, g/d	4			3			3			-2			-3			-5			-2			-2.88		
Slope bias, g/d	0.315			0.55			0.55			0.15			0.16			0.19			0.15			0.16		
P _{Mean bias}	0.38			0.53			0.58			0.64			0.56			0.31			0.64			0.53		
P _{Slope bias}	0.0001			0.0001			0.0001			0.0001			0.0001			0.0001			0.0001			0.0001		

¹σ² = mean random effect variance; τ₀₀ = between group variance; AIC = Akaike information criterion; BW = BW centered to the mean value of 612 kg; CCC = concordance correlation coefficient; conditional R² represents fixed and random effects variance; DEIn = digested energy intake; DEInp = DEI minus DE from MPIn; EAA = essential AA intake; ICC = interclass correlation coefficient; marginal R² represents the fixed effects variance; MPIn = MP intake; Obs = observed; Pred = predicted; RMSE = root mean squared error (g/d or % of Obs mean); mean and slope bias % expressed as a % of the MSE.

DEInp increases (NASEM, 2021). Therefore, the efficiency of conversion at maintenance intake would be expected to be less than at production level intakes due to the greater energy density in diets for lactating cows as compared with dry cows. Additional work is needed to understand this inconsistency. Currently, the factorial method is used to represent nonproductive use which includes metabolic fecal protein, and a regression equation with a much smaller intercept is used to represent MP use for milk protein.

Adding BW centered to the mean of the data to account for different sized animals (Equation [3.6]) resulted in a slope coefficient of -0.35 g/kg BW, a 13 g/d decline in the intercept, no real changes in the other slope coefficients, and no improvement in overall fit statistics. The lack of an improvement in fit statistics with the inclusion of BW in the regression would normally indicate the added complexity should be excluded, however, as the model should be as widely applicable as possible, including for small breeds such as Jerseys, and the latter are very poorly represented in the development database, it seemed prudent to retain the term given its significance and general biological consistency. At a BW of 700 kg (Holstein breed average; NASEM, 2021), the nonproductive NP requirement estimate from Equation [3.6] is 209 g NP/d for a fasting, nonpregnant, nonlactating cow as compared with an estimate of 241 g/d for the same cow by the NASEM (2021) model. Equation [3.6] predicts a requirement for a 520 kg Jersey in the same state as 146 g/d, and the NASEM (2021) model as 180 g/d. The differences between the high and low BW cows calculated by the 2 models are almost identical at 32 and 34 g/d, respectively, indicating the BW term of Equation [3.6] was justified and appropriately represented.

The linear effects of energy intake and the curvilinear responses to protein are consistent with declining marginal returns to MP supply as supply increases (Whitelaw et al., 1986; Hanigan et al., 1998; Kalscheur et al., 2006; Arriola Apelo et al., 2014b; Daniel et al., 2016), which presumably reflects saturation of the capacity of the mammary tissue to synthesize milk protein at a given energy supply. However, the latter responses are within a given set of animals or cells whereas the overall response captured by a squared MPIn term represents a global response to MPIn across the experimental population of animals. The apex of MPIn responsiveness was predicted to occur at 3,352 g MPIn/d from the linear and squared coefficients for Equation [3.5]. This yields a maximum predicted milk protein of 1,511 g/d. Thus animals with low potential to produce milk protein, such as the cows of Whitelaw et al. (1986), would be predicted to exhibit linear responses to MPIn given low MPIn intakes while very high merit animals would be predicted to operate above the apex predicted with this equation. Although

the squared term captures at least part of the nonlinear responses in the cows within this dataset, it cannot be expected to be globally applicable to the full genetic range of animals because those of high genetic merit will have greater synthetic capacities than represented by the data herein and the resulting equation. Conversely, the cows of Whitelaw et al. (1986) did not exhibit linear responses because their genetic capacity was low.

We also observed independent quadratic responses to RDP and RUP intakes when replacing MPIn in the equation. The slope coefficients were, however, seemingly greater for both than reported in NRC (2001) at 0.34 for RDP intake g/g and 0.24 g/g for RUP intake, likely because the prior estimates would have been colinear with the DMI term used in the model. At a mean DMI of 20.7 kg/d, these estimates equate to a zenith for milk protein at 15.9% RDP and 11.2% RUP, which are both greater than observed by NRC (2001), but are consistent with the generally linear response of microbial protein to RDP (NASEM, 2021) and milk protein to MPIn herein.

(Orskov, 1992) hypothesized that maximum protein synthesis was related to energy supply to the tissue. As energy is tightly regulated, the partial efficiency of energy use for milk production has been constant over time (Moraes et al., 2015), thus increases in genetic gain in milk production are associated with increased energy intake. If energy supply is related to maximum rates of protein production, this would present as an interaction between energy and protein terms as observed by Hanigan et al. (1998). Unfortunately, we were not able to identify any interactions between DEInp and MPIn including the form used by Hanigan et al. (1998). Any addition of higher order DEInp terms resulted in non-significant estimates for those terms, and an increase in AIC. These findings are consistent with the lack of interactions found at the cellular (Clark et al., 1978; Appuhamy et al., 2011b; Appuhamy et al., 2014a) and animal levels (Vanhatalo et al., 2003; Raggio et al., 2006; Rius et al., 2010c). Although there must be a maximal rate of signaling and protein synthesis that can occur, the energy and hormonal arms of the signaling pathways do not appear to dictate that plateau within normal physiological ranges (Arriola Apelo et al., 2014c,d; Castro et al., 2016a), and energy supply cannot be used to scale maximum protein production to accommodate very high levels of production observed on North American dairy farms. It remains to be seen whether mathematical scaling based on pen or herd production will suffice (NASEM, 2021).

Replacing DEInp with DE concentration in DM (data not shown) resulted in much poorer predictions as compared with Equation [3.5] with a non-significant intercept. Attempts to express MPIn as a ratio to DEIn, as in some swine models (NRC, 2012), also resulted in fits that were poorer than Equation [3.5] based on AIC. The

latter approach requires careful exploration of the ratio to ensure it is normally distributed and it is linearly related to the dependent variable (Lien et al., 2017). Lien et al. (2017) reminds us that if the ratio is linearly related to the dependent variable, its reciprocal must be nonlinearly related, and a nonlinear relationship with the independent variable violates the linear regression assumptions. Use of a ratio also forces a fixed relationship between the 2 variables rather than letting the relationship be discovered via combinations of linear and higher order terms for each.

Models Representing the Effects of Individual EAA

A summary of evaluations of 4 different methods of expressing the effects of individual EAA on milk protein synthesis (Equations [3], [4], [5], and [6], each reduced by backward elimination) is provided in Table 4 (Equations [4.1] to [4.4]). As for the MPIn-based models, DEInp was found to be highly significant for all of the models, as were the partitioned energy terms (DEStIn, DENDFin, DEFAIn, and DEROMIn; Mcal/d). In no case, did the effect of systemic infusions (primarily jugular or mesenteric vein infusions) remain in the model after the selection process suggesting that losses during absorption from dietary sources were not significantly greater than from direct systemic infusions given the errors of measurement.

Absorbed EAA Expressed as Grams per Day with Individual EAA Quadratics. Reduction of Equation [3] yielded Equation [4.1] which had linear terms for DEStIn, DEROMIn, DENDFin, DEFAIn, Arg, His, Lys, Phe, Thr, Trp, and Val, and quadratic terms for Arg, Met, Phe, Thr, and Trp (Table 4). The mixed effects model had an AIC of 10,360 and a conditional R^2 of 0.87. The fixed effects portion of the model had a marginal R^2 of 0.51, and RMSE of 131 g/d, which was 14.2% of the observed mean, no mean bias, and only 4% of the mean squared error (MSE) associated with a slope bias. As the mixed model was fitted to the data, the latter reflects some structured random effects not captured by the fixed effects terms. This equation form has generally been used in the past as it was thought to approximate the saturation effects of each EAA, and the curves could be linearized to identify a breakpoint which was deemed the requirement (NRC, 2001). However, the negative linear responses for Thr and Val are difficult to explain biologically. The negative effect of Val may be an indication of an antagonism of Ile and Leu, although neither were present in the final model. If Val were truly antagonistic, one would expect a positive slope for at least one of the other 2 branched-chain AA (BCAA). It is also possible that the result is due to co-linearity among variables. The correlations between absorbed Val and Ile and Leu were high at 0.90 and

0.92, although several treatments deviated significantly in the relationship between Leu and Val (Supplemental Figure S1). There was less variation between Ile and Val, but the low correlations among the linear parameter estimates from the regression model (generally less than |0.10| with only His exceeding that range at -0.38) do not support that argument (data not shown). The response might also be due to lack of independent variation from total EAA with Val having the highest correlation at 0.99 (Supplemental Figure S1). Thus, the source of the negative response is unclear, but it seems likely to reflect an autocorrelation problem, and thus is biologically suspect.

The presence of a negative linear term for Thr is even more difficult to explain. The zenith of the quadratic occurs at an absorbed Thr supply of 82 g/d with a partial contribution to milk protein of -759 g/d. The partial contribution only reaches 0 at just under the maximum prediction for absorbed Thr, and thus it has a large negative effect throughout the expected dietary range. This is the opposite of that observed by Arriola Apelo et al. (2014d), and thus the response is not supported by direct experimental evidence at the tissue level. But Thr has the 2nd highest correlation with the total absorbed EAA at 0.98, and correlations among parameter estimates were generally greater for Thr with the correlation between Thr and Phe being -0.63. This may explain the relatively large positive slopes for Arg and Phe. Therefore, the negative linear slope may also represent an auto-correlation problem rather than a biological response. Finally, all 4 of the EAA having linear and quadratic terms exhibited very high correlations (< -0.97) between the linear and quadratic terms within an EAA. This is a clear indication that the higher order terms were not independently defined by the data, and thus contributing to the overall model definition problem, and likely the cause of the negative linear term for Thr.

Responses to Phe were predicted to plateau at 103 g/d with a partial contribution to milk protein of 275 g/d. The partial contribution of Trp to milk protein was 280 g at the minimum absorbed supply of 9 g/d (51% gross Trp efficiency based on milk protein composition of NASEM, 2021) and plateauing at 346 g/d with an absorbed supply of 21 g/d (27% gross Trp efficiency). The partial contribution of absorbed Met to milk protein was predicted to be 27 g/d at the mean absorbed Met value of 46 g/d, yielding a partial response of 0.59 g milk protein/g absorbed Met for a mean gross Met efficiency of 1.8%. Such a low response is not supported by the direct experimental evidence of Pisulewski et al. (1996) who observed a linear response from 1,010 to 1,100 g milk protein to infused Met ranging from 0 to 24 g/d. This reduces to a slope of approximately 3.75 g milk protein/g absorbed Met, and equates to a gross absorbed Met efficiency of 11.3%.

Table 4. Evaluation of the form of expression of the individual EAA; models were derived by backward elimination from the global models (Equations [3] to [6]) until all terms were significant at $P < 0.05$ ¹

Term ⁶	Equation [4.1] ²		Equation [4.2] ³		Equation [4.3] ⁴		Equation [4.4] ⁵	
	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	−35	73	−138	47	−3687	387	−480	104
BW − 612, kg	−0.323	0.14	−0.346	0.136	−0.413	0.13	−0.375	0.14
Arg	8.23	2.1			212	51	287	116
His	1.70	0.75	2.40	0.79	261	47		
Ile			1.73	0.53				
Leu								
Lys	0.823	0.25	1.30	0.27	76.6	14.8	39.0	12
Met			1.70	0.40	107	23.5		
Phe	6.01	2.1			12.8	5.1	242	100
Thr	−17.6	3.8	2.60	0.88	17.7	8.2	−696	204
Trp	36.8	11.8	−6.75	2.1	657	159	1616	666
Val	−1.46	0.61						
EAA					0.459	0.029		
NEAA			0.130	0.053				
Arg ²	−0.0316	0.0090			−9.97	2.4	−54.3	23
His ²					−26.0	5.0		
Lys ²					−2.25	0.50		
Met ²	0.0129	0.0036			−8.42	2.1	26.0	8.1
Phe ²	−0.0290	0.0091					−52.3	19
Thr ²	0.107	0.020					189	47
Trp ²	−0.889	0.24			−150	34	−1,733	603
Σ(EAA ²)			−0.00169	0.00030				
DEStIn	10.1	0.82	10.5	0.83	9.56	0.82	19.5	0.81
DErOMIn	10.7	1.5	11.9	1.5	11.0	1.4	20.3	1.4
DENDfIn	5.09	1.2	5.00	1.2	4.96	1.2	15.1	1.2
DEFAln	9.48	1.8	10.1	1.8	9.37	1.8	18.2	1.9
AIC	10,360		10,412		10,363		10,412	
σ ²	6,079		6,643		6,153		6,674	
τ ₀₀ (PubID)	16,292		16,120		15,727		15,601	
ICC	0.73		0.71		0.72		0.70	
Marginal R ²	0.51		0.50		0.52		0.51	
Conditional R ²	0.87		0.85		0.87		0.85	
Obs mean, g/d	922		922		922		922	
Pred mean, g/d	925		927		926		925	
CCC	0.76		0.76		0.77		0.76	
RMSE, g/d	131		132		129		129	
RMSE, %	14.2		14.3		14.0		14.0	
Mean bias, %	0.1		0.1		0.1		0.1	
Slope bias, %	4.0		4.2		3.3		4.6	
Mean bias, g/d	−3		−5		−4		−3	
Slope bias, g/g	0.172		0.178		0.151		0.182	
P _{Mean bias}	0.47		0.29		0.35		0.44	
P _{Slope bias}	0.0001		0.0001		0.0001		0.0001	

¹Data (n = 905) were weighted by the square root of the number of experimental units per treatment and experiment ID was included as a random variable.

²Derived from Equation [3]. AA expressed as g/d.

³Derived from Equation [4]. AA expressed as g/d.

⁴Derived from Equation [5]. AA expressed as % of EAA.

⁵Derived from Equation [6]. AA expressed as g/Mcal.

⁶σ² = mean random effect variance; τ₀₀ = between group variance; AIC = Akaike information criterion; conditional R² considers fixed plus random effects variance; EAA = total absorbed EAA; individual absorbed EAA terms are denoted by the standard AA abbreviations and expressed as g/d for Equation [4.1] and [4.2], as % of EAA for Equation [4.3], and as g/Mcal nonprotein digested energy intake for Equation [4.4]; DEInp = digested energy intake minus the DE from absorbed true protein (Mcal/d); DEFAln = digested energy (DE) from fatty acid intake (Mcal/d); DENDfIn = DE from NDF intake (Mcal/d); DErOMIn = digested energy intake from residual OM (Mcal/d); DEStIn = DE from starch intake (Mcal/d); ICC = interclass correlation coefficient; marginal R² represents the fixed effects variance; Obs = observed; Pred = predicted; RMSE = root mean squared error (g/d or % of Obs mean); mean and slope bias % are expressed as a % of the MSE.

Plateau values were not observed for His, Lys, or Met offering no information on target requirement levels under the classical approach. From the quadratics, plateaus

for Arg, Phe, Thr, and Trp occur at 130, 104, −82, and 21 g absorbed/d, respectively. Obviously the Thr response and plateau are non-biological. Arriola Apelo et al. (2014d)

observed that in vivo concentrations fell quite low on the linear portion of the quadratic curves, thus, the lack of quadratic effects for some EAA was likely due to lack of range in the in vivo data.

Of the terms in the reduced model, the linear effects of His, Trp, and Val were marginally defined by the data having CV of 59, 49, and 49% respectively with SD of the CV of 77, 20, and 27%, respectively under exhaustive cross-evaluation (data not presented). Surprisingly, the quadratic terms were all fairly stable despite the high correlations with the linear terms. Regarding His, Trp, and Val, the large CV are indicative of the parameter estimates being a function of a relatively small portion of the overall data, which minimally suggest more data are required, but it could also be an indication of heterogeneity in the responses for some datasets.

Although Equation [4.1] performed relatively well on cross-evaluation, suggesting that the solution was stable, the large linear slope values for Arg, Phe, and Trp; the negative linear slopes for Thr and Val; the exponential response for Met; and the high correlations among some parameters suggests the model was overparameterized and likely fails to capture the true response, which argues against its use for field application.

Absorbed EAA Expressed as Grams per Day with a Common EAA Quadratic. To address the apparent overparameterization problem of Equation [4.1], and recognizing that at least mTOR signaling is integrated across AA, energy, and hormonal signals yielding a common regulatory response (Castro et al., 2016a; Sabatini, 2017), we hypothesized that protein synthesis capacity could also be represented with a combined saturation point. Thus the individual quadratic terms for EAA in Equation [3] were replaced with a common quadratic slope term applied to the sum of the individual squared EAA yielding Equation [4] (Table 4). This form has the implicit assumption that the saturation response to individual EAA is additive and the responses to each EAA are similar. It is important to point out that both Equations [3] and [4] also imply that AA are interchangeable from a regulatory standpoint, and a single AA at very high concentrations can substitute for low concentrations of other AA. This could lead to mass balance problems; however, the substantial regulation of mammary AA transport mitigates the potential for such a problem (Arriola Apelo et al., 2014b), and based on the observed efficiencies of conversion of individual EAA to milk protein predicted from the system (see below), this does not seem to occur within in vivo ranges. Arriola Apelo et al. (2014d) used a similar set of quadratics for his work and found that the plateaus occurred at EAA concentrations well above physiological ranges. As non-linearity with respect to DEInp was not observed when exploring the effects of

DEInp and MPIn, it was not considered a contributor to the quadratic term.

Correlations and covariance among the fitted parameters was assessed for the Equation [4], and the results are presented in Supplemental Table S2 (see Notes). The correlations among parameters were surprisingly low having a maximum in absolute terms of 0.53 with most below absolute values of 0.30, indicating the model was generally uniquely definable from the data.

As before, Equation [4] was reduced by backward elimination resulting in Equation [4.2]. The reduced equation had 4 fewer terms, an AIC of 10,412 and a conditional R^2 of 0.85. The fixed effects portion of the model had a marginal R^2 of 0.50, and an RMSE of 132 g/d, which was 14.3% of the mean, no mean bias, and only 4.2% of the MSE associated with a slope bias. Based on AIC, which has a modest penalty for the number of parameters in the model, the mixed model performed less well than Equation [4.1]; however, the RMSE for the fixed components of the model was only 1 g/d greater than for the prior equation, with no real differences in mean or slope bias and a marginal R^2 of 0.50 vs 0.51 for the prior, indicating no real difference in performance between the models when random effects are unknown, as is the case for field application to novel groups of animals and diets.

Equation [4.2] had positive linear terms for all 4 energy coefficients plus His, Ile, Lys, Met, Thr, and the NEAA. Contrary to Equation [4.1], the linear slope coefficient for Trp was highly negative, adding further to the question of whether the data are adequate to uniquely define Trp effects.

The quadratic term, applied to the sum of the individual squares of each EAA, was highly significant, further supporting the need for some form of nonlinearity in the model. From the quadratics, absorbed EAA inputs (g/d) at response plateaus were: His = 710, Ile = 512, Lys = 385, Met = 503, Thr = 769, and Trp = -1997. These are 1,480, 440, 250, 1,090, 590, and -7,680% of the mean inputs of each (Table 1), respectively. Comparison of the casein synthesis plateaus observed by Arriola Apelo et al. (2014d) with mean plasma AA concentrations reported by Swanepoel et al. (2016) yields approximate plateaus relative to mean concentration of 300% for Ile, 200% for Leu, 800% for Met, and 300% for Thr. Thus both observations suggest the plateaus for individual EAA are far above observed plasma concentrations yielding mostly linear responses in the normal in vivo range which explains the difficulty in deriving individual quadratic terms for each EAA.

The NEAA term was also significant suggesting that, although a small effect, the NEAA play some role in driving milk protein output and thus should not be totally ignored. This is consistent with observations of sestrin-2

signaling in response to EAA and NEAA (Luo et al., 2018), which would impact mTOR phosphorylation and protein synthesis. Several NEAA are also actively transported into cells and subsequently exchanged for extracellular EAA via the LAT1, LAT2, Y⁺LAT1, and Y⁺LAT2 transporters (Broer and Broer, 2017). Thus, low blood concentrations of NEAA may reduce EAA transport activity (Hruby Weston et al., 2024) leading to the reduction in milk protein output.

Excepting Trp, the individual EAA terms were more consistent across EAA. Using the milk protein composition of NASEM (2021), the gross efficiencies of conversion of absorbed EAA to milk protein EAA based solely on the linear terms were His = 7.0%, Ile = 10.7%, Lys = 11.5%, Met = 5.2%, and Thr = 12.0%.

Unlike the EAA terms, the DE terms in both equation forms were fairly consistent with the DE from starch, rOM, and FA all having similar coefficients of approximately 10 g milk protein/Mcal of DE, and the DE from NDF having a coefficient of 5 g milk protein/Mcal. The lesser effect of DENDF_{In} likely reflects the lower potentiation of insulin and IGF-1 secretion as compared with digested starch and rOM (Garnsworthy et al., 2009), and it is consistent with the negative digested NDF term in the milk protein equation of NASEM (2021). The NASEM (2021) equation used an aggregated DE_{Inp} term for energy where starch, rOM, FA, and NDF were given equal weight. As digested NDF apparently has half the effect, the regression solved for a negative term for digested NDF concentrations to offset a portion of the digested NDF response within the DE_{Inp} term.

The effect of DE from digested FA was presumably mediated via energy supply to the mammary glands as dietary FA do not generally drive ruminal microbial growth (Hanigan et al., 2021) or insulin and IGF-1 concentrations (Becú-Villalobos et al., 2007), although highly saturated FA have been observed to elicit an insulin response (Harvatine and Allen, 2005).

The intercept was more negative than for Equation [4.1] and similar to that derived for MP_{In} (Equation [3.5]), although less than expected based on factorial summations of NP requirements summarized by NASEM (2021). If we assume the factorial estimates are unbiased, an intercept of -138 g NP/d may indicate a slight potential for slope bias in the milk protein predictions at very low EAA inputs.

The correlations among the linear EAA terms and the quadratic term ranged from a low of -0.12 for Thr to -0.41 for His. Based on these, one can conclude that the slope of the common term was generally independently defined by the data.

The model exhibited slightly greater parameter stability during cross-evaluations (data not presented) with only the NEAA term exhibiting some uncertainty with

a CV of 48.2%. The Trp term was more stable with a CV of 33.7%, and exhibited maximal correlations with Lys and Thr terms at modest values of -0.40 and -0.36, respectively. Thus, this simpler model resulted in slightly more stable slope estimates that were consistent with the known biology excepting the Trp slope. Based on these attributes, the similar fixed effects performance favors Equation [4.2] over Equation [4.1].

Interactions Among Nutrients

Based on prior work showing that interactions among nutrients was not generally significant (Vanhatalo et al., 2003; Raggio et al., 2006; Rius et al., 2010c; Appuhamy et al., 2011b, 2012, 2014a; Arriola Apelo et al., 2014d; Yoder et al., 2020), our screening work was conducted without consideration of potential interactions. To test this assumption, we examined all the potential 2-way interactions among the linear energy and EAA terms in Equation [4]. Reduction of the modified equation by backward elimination resulted in Equation [S3.1] (Supplemental Table S3, see Notes), which contained 20 significant, 2-way interactions. Although the model performed slightly better than Equations [4.2] and [4.1] (AIC = 10,296, marginal R² of 0.54, and an RMSE of 128 g/d [13.9%]), an RMSE gain of 4 g/d hardly justifies the addition of 20 parameters and the associated possibilities for co-linearity problems and spurious estimates driven by outliers. Additional discussion is provided in the Supplemental Material: Interactions Among Nutrients (see Notes), and no further consideration was given to the approach.

Absorbed EAA as a Proportion of Total EAA. Equation [5] represents an equation form with linear terms for DE and total EAA supplies, and linear and quadratic terms for the percentages of total EAA represented by each EAA. The subcomponents of DE_{Inp} and EAA supply captures the independent effects of energy and protein and ensures mass balance. The remaining AA terms reflect the effects of the EAA composition of the total EAA supply on milk protein production, and as for Equation [3], independent plateaus are implied by the individual quadratics for each EAA. As before, the model was reduced by backward elimination resulting in Equation [4.3], which performed similar to Equation [4.1] and slightly better than Equation [4.2] based on an AIC of 10,363. The model had 19 fixed terms as compared with 14 terms for Equation [4.2]. The RMSE for the fixed components was 129 g/d which was marginally better than Equations [4.1] and [4.2] by 3 g/d. The relative prediction error was 14.0% of the observed mean, with no mean bias and minimal slope bias.

The linear terms for absorbed Phe and Thr were very unstable during cross-evaluation (data not presented) having CV of 403 and 59%, respectively. This is an

indication that the data were not adequate to define the responses given the variation in responses among trials. In the absence of additional observational data, they should be removed from the model. All other terms had CV less than 36% indicating relative stability during cross-evaluation.

The correlation matrix suggested some collinearity problems with negative correlations among the linear and quadratic absorbed EAA terms ranging as great as -0.98 . There were also moderate correlations of 0.50 for DEStIn and DENDFin, -0.48 for Phe and the quadratic Trp term, and 0.47 for absorbed Phe and Trp percentages. The remainder were below absolute values of 0.43 with most less than 0.30 . Removal of the Arg, Phe, Thr, and Trp quadratic terms addressed the collinearity problems, but the AIC increased to $10,436$.

The intercept was $-3,687$ g/d, which greatly exceeded the mean MPIn and even CPIn, and thus was well outside of the biologically possible range. The model is clearly centered in the lactating data used to fit the model, but it must transition from the center of the data to a biologically impossible solution as the inputs approach 0 , and thus predictions from low inputs would be highly suspect. When the quadratic terms were reduced to address collinearity concerns, the intercept assumed a value of 77 g/d, which was greater than expected, but greatly improved over Equation [4.2].

Estimates for the effects of the DE components were essentially the same as for the other models, and the linear slope for total EAA was 0.459 g milk protein/g EAA which was less than for Equation [3.8] indicating the individual quadratic terms for the EAA proportions did not exert as much influence as the summative term in Equation [3.8]. There were positive linear effects for the percentages of Arg, His, Lys, Met, Phe, Thr, and Trp; and negative quadratic effects for Arg, His, Lys, Met, and Trp. The plateaus for each EAA were: Arg = 10.6% , His = 5.0% , Lys = 17.0% , Met = 6.3% , and Trp = 2.2% . With the EAA representing an average of 44.8% of the total AA supply, these can be converted to the equivalent of percentages of MP yielding: Arg = 4.77% , His = 2.25% , Lys = 7.63% , Met = 2.84% , and Trp = 0.98% , respectively. The Lys and Met zeniths compare favorably with the estimates derived by NRC (2001).

Excepting the linear Phe and Thr terms and the collinearity concerns with the quadratic terms, this model was fairly well defined by the data and consistent with prior requirement models, but the form of expression of the individual EAA responses is problematic. Because each term is divided by total EAA, a change in the absorbed supply of any of the individual EAA not only causes a change in the numerator for that EAA, it also causes a change in the denominator for all other EAA. This results in nonlinearity in the relationships among the ratios and

the independent variable (a violation of the linear regression assumptions), and heterogeneity in individual EAA slopes with the contribution of each EAA to milk protein varying depending on the summative values of the other EAA. Thus, the effect of the addition of 1 g of absorbed Met will be different when added alone than when added with another EAA or as a component of a protein. In the latter case, the effect will be quite different given the much larger change in the denominator. Such a response is not consistent with existing knowledge where each factor primarily exerts an independent effect (Rius et al., 2010c; Appuhamy et al., 2011b; Appuhamy et al., 2012; Appuhamy et al., 2014a; Arriola Apelo et al., 2014d; Yoder et al., 2020). However, the ability of the mammary glands to alter individual AA transport activity in response to AA supplies (Bequette et al., 2000; Hanigan et al., 2000), and the flexing of liver clearance in the opposite direction yields variable efficiency of conversion of AA supply to milk protein (Fleming et al., 2019a), which is perhaps the source of the interactions identified in Supplemental Table S3. Such flexing across the tissue beds, which generally reduces mammary affinity and increases splanchnic affinity as absorbed supplies increase, is directionally consistent with the denominator effects within the ratios of Equation [5] and its reduced forms (Table 4), but the latter are inconsistent with the 9 positive, 2-way interactions among AA identified in Supplemental Equation [S3.1], and the ratios are highly unlikely to replicate in vivo response surface given the ratios are fixed mathematically. Finally, the positive interactions identified in Supplemental Table S3 are directionally and quantitatively inconsistent with the negative effects generated by the encoded ratios. For these several reasons, Equation [4.3] cannot be recommended for general field use, and was not considered further.

Absorbed EAA Expressed as a Ratio to Nonprotein DE Intakes. The final equation form that was considered expressed the effect of individual absorbed EAA as a ratio to DEInp (Equation [6]). This form is used in swine models (NRC, 2012). It suffers from the same mathematical deficiencies and violation of linearity assumptions as Equation [5] due to the use of the ratios. Backward elimination yielded Equation [4.4], which had 16 fixed terms. There were positive linear terms for the DE components, Arg, Lys, Phe, and Trp and a negative linear term for Thr. It had negative quadratic effects for Arg, Phe, and Trp, and positive quadratic effects for Met and Thr. The AIC was $10,412$, which was the same as Equation [4.2] and greater than the other equations. The marginal R^2 was 0.51 , and the RMSE was 129 g/d with no meaningful mean or slope bias, and a relative prediction error of 14% of the observed mean.

Under cross-evaluation, the linear and quadratic Trp terms and the quadratic Arg term were unstable with

CV of 96%, 49%, and 49%, respectively. The quadratic terms for Arg, Phe, Thr, and Trp were highly correlated with the respective linear terms at -0.99 for all indicating inadequate data range to define those terms. The Thr quadratic term was also correlated with the Trp and Phe quadratic terms at -0.61 and -0.60 , respectively. There were also correlations among the linear terms for Phe, Thr, Arg, and Trp all exceeding -0.58 . There were cross-correlations among the linear EAA terms and other quadratic terms ranging from 0.55 to 0.60 . Removal of the quadratics for Arg, Phe, Thr, and Trp resolved all of the correlation problems, but the AIC increased to $10,443$ indicating the model was clearly inferior to the other forms. Based on inferior performance, the wide range in collinearity among the parameters, parameter stability problems during cross-evaluation, and the use of ratios in the model, the model cannot be recommended for field application, and was not considered further.

Model Selection by Evaluation of All Possible Models

The screening work established that Equation [4] was generally biologically consistent, excepting the negative linear Trp slope, and the data were adequate to define the energy terms and at least 5 EAA. Because backward and forward elimination algorithms can result in biased model selection (Heinze and Dunkler, 2017), we evaluated all possible combinations of the fixed terms in Equation [4]. A summary of the top 5,000 models based on AIC from a total population of $524,288$ is provided in Supplemental Table S4 (see Notes). DEFAIn, DEROMIn, and DESTIn were in all of the top 5,000 models, and DENDFin was in $4,554$ of the $5,000$ underscoring the importance of DE as a driving factor for milk protein production. Absorbed Lys, Met, and the quadratic EAA term were in $4,535$, $4,718$, and $4,057$ of the $5,000$ indicating their relative importance in explaining variation in milk protein production. His and Ile were in more than $3,200$ of the models, and the remaining AA including the aggregated NEAA were in roughly half of the top models. Parity and BW were in $2,271$ and $3,180$ of the top models, respectively. From the CV for the parameter estimates for the $5,000$ models, it can be observed that several of the potential terms had extremely variable slope estimates, indicating they were heavily dependent on the presence or absence of small subsets of the data, and thus not well defined. These included absorbed Arg, Leu, Phe, and Val. The remaining terms were more consistent across models indicating they were well defined by the data, and should represent unbiased estimates of the true effects.

Table 5 (Equations [5.1] to [5.7]) contains a summary of a selection from the top models including the global model (Equation [5.1]) and the weighted average model

(Equation [5.2]). The best model (Equation [5.3]) contained linear terms for absorbed His, Ile, Lys, Met, Thr, Trp, and NEAA; all 4 DE terms; and BW. It had an AIC of $10,389$ which was 8 units less than the global model. The difference in AIC is due to the penalty applied to the number of terms in the model. Although this model was the best based on AIC, it lacked some biological coherence given the negative slope for absorbed Trp. The global and weighted average models also had negative terms for Trp.

The first model that did not include a negative linear slope for Trp was rank order 167 (Equation [5.4]). It had an AIC of $10,397$, a marginal R^2 of 0.49 , and RMSE of 132 g/d, a relative error of 14.3% , and negligible mean and slope bias. In this regard, it performed almost identically to the top and global models, and had the advantage of biological consistency. The intercept was numerically positive, but not significantly different from 0. The absence of the relatively large negative Trp term resulted in a general reduction in the slope estimates for the other EAA except for Met. Given the low correlations among parameter estimates, the changes should not be due to co-linearity problems. It is more likely the general reductions in slope estimates simply reflect the removal of the significant negative driving term, which algebraically requires an offsetting reduction in one or more other parameters (all positive) to allow the predictions to remain centered in the data. Although the negative slope for Trp is seemingly biologically inconsistent, it clearly is capturing some aspect of the observed variance.

The derived slopes of 0.96 g milk protein/g absorbed Lys and 1.92 g milk protein/g absorbed Met were well below the ranges of 16 to 4 g milk protein/g Met intake and 5.0 to 3.2 g milk protein/g Lys intake reported by Vyas and Erdman (2009). However, Vyas and Erdman (2009) only controlled independently for Lys and Met intake. Any DMI or DE concentration changes that occurred would have altered intakes of DE and the other EAA, and the effects of those would have been assigned to Lys and Met. This is a potential problem in any work that does not consider the full range of potential factors contributing to the biological function of interest. Lean et al. (2018) reported coefficients of 1.04 g milk protein/g metabolizable Lys and 1.89 g milk protein/g metabolizable Met, which are essentially equivalent to values derived using Equation [5.4] and other models where negative linear EAA terms were excluded. Plots of residual errors for milk protein production also support these lower estimates given essentially no evidence of bias across the entire dataset (Figure 3 and Figure 4), studies using infusions (Supplemental Figure S2), or studies feeding rumen-protected Lys and Met (Supplemental Figure S3) with respect to the supply of any of the EAA. The lack of a slope for the residuals within

Table 5. Selected results from evaluation of all possible models derived from Equation [4]; rank order was based on AIC; data (n = 905) were weighted by the square root of N and publication ID was included as a random variable

Term ²	Equation [5.1]		Equation [5.2]		Equation [5.3]		Equation [5.4]		Equation [5.5]		Equation [5.6]		Equation [5.7]	
	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	48.3	99	68.2	108	75.1	88	115	80	55	98	57	98	124	88
BW – 612	–0.378	0.14	–0.362	0.14	–0.348	0.14	–0.357	0.14	–0.353	0.14	–0.354	0.14	–0.349	0.14
Arg, g/d	0.55	0.54	0.56	0.56										
His	2.76	0.89	2.58	0.90	2.41	0.78	1.78	0.76	1.77	0.76	1.71	0.76	2.24	0.72
Ile	1.74	0.61	1.76	0.60	1.73	0.53	1.34	0.52	1.34	0.52	1.29	0.52	1.75	0.47
Leu	0.38	0.31	0.31	0.32	1.30	0.26	0.96	0.24	0.37	0.26	0.36	0.26		
Lys	1.37	0.30	1.26	0.30	1.70	0.40	1.92	0.40	1.20	0.30	1.18	0.30	1.12	0.23
Met	1.69	0.41	1.73	0.41	1.70	0.40			1.83	0.40	1.81	0.40	1.90	0.40
Phe	–0.003	0.435	–0.004	0.441										
Thr	2.86	0.90	2.81	0.93	2.57	0.87	1.59	0.82	1.76	0.82	1.82	0.83		
Trp	–6.15	2.47	–6.40	2.39	–6.73	2.11								
Val	–0.852	0.784	–0.953	0.821										
NEAA	0.097	0.058	0.121	0.057	0.132	0.053	0.114	0.053	0.117	0.053	0.120	0.053	0.153	0.049
lnTP _{Art}	0.161	0.178	0.173	0.180										
Σ(EAA) ²	–0.00195	0.00048	–0.00164	0.00043	–0.00169	0.00030	–0.00145	0.00029	–0.00193	0.00045	–0.00192	0.00045	–0.00143	0.00029
DEFAIn	10.2	1.8	10.2	1.8	10.2	1.8	10.3	1.8	10.2	1.8	10.2	1.8	10.3	1.8
DENDIn	5.22	1.25	5.17	1.27	5.01	1.24	5.05	1.25	5.07	1.24	5.08	1.24	5.23	1.24
DEROMIn	12.1	1.5	12.0	1.5	11.9	1.5	11.7	1.5	12.0	1.5	12.0	1.5	12.0	1.5
DESTIn	10.5	0.8	10.6	0.8	10.6	0.8	10.7	0.8	10.6	0.8	10.7	0.8	10.8	0.8
Parity – 1	28.0	26.6	26.0	26.9										
AIC	10,397		NA		10,389		10,397		10,397		10,398		10,399	
σ ²	6,604				6,652		6,730		6,713		6,709		6,748	
τ ₀₀ (PubID)	16,172		16,172		16,167		16,321		16,310		16,196		16,472	
ICC	0.71		0.71		0.71		0.71		0.71		0.71		0.71	
Marginal R ²	0.50		0.50		0.50		0.49		0.50		0.50		0.49	
Conditional R ²	0.856		0.854		0.854		0.852		0.853		0.853		0.851	
Predict mean, g/d	927		927		927		926		926		926		926	
CCC	0.76		0.76		0.76		0.75		0.75		0.76		0.75	
RMSE, g/d	132		131		132		132		132		132		133	
RMSE, % of mean	14.3		14.2		14.3		14.3		14.3		14.3		14.4	
Mean bias, % of	0.1		0.1		0.1		0.1		0.1		0.1		0.1	
MSE														
Slope bias, % of	4.1		4.3		4.2		4.4		4.2		4.3		4.6	
MSE														
Mean bias, g/d	–5		–4		–4		–4		–4		–4		–4	
Slope bias, g/g	0.175		0.180		0.179		0.184		0.180		0.182		0.191	
P _{Mean bias}	0.27		0.31		0.31		0.37		0.35		0.35		0.36	
P _{Slope bias}	0.0001		0.0001		0.0001		0.0001		0.0001		0.0001		0.0001	

¹Rank order of the models based on AIC where the population of models represented all possible combinations of the terms. Average = averaged coefficients across the top 5,000 models.

²AIC = Akaike's information criterion; BW = body weight (kg); conditional R² represents fixed and random effects variance; DEFAIn = digested energy intake from fatty acids (Mcal/d); DENDIn = digested energy intake from NDF (Mcal/d); DEROMIn = digested energy intake from residual OM (Mcal/d); DESTIn = digested energy intake from starch (Mcal/d); lnTP_{Art} = the effect of systemic infusion of AA versus intestinal absorption; ICC = interclass correlation coefficient; marginal R² considers only the fixed effects variance; RMSE = root mean squared error (g/d or % of observed mean); σ² = mean random effect variance; τ₀₀ = between group variance; NA = not available.

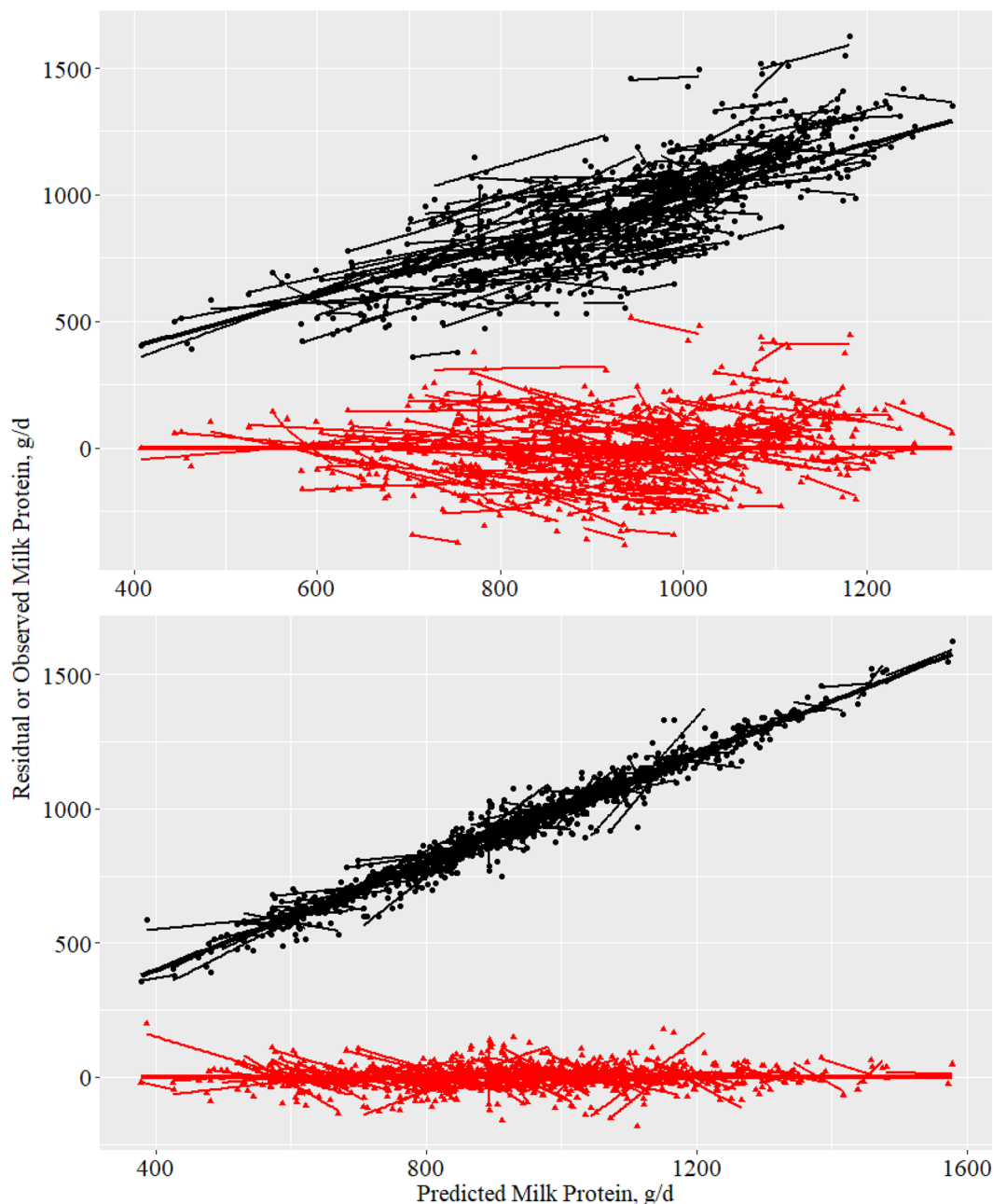


Figure 3. Residual errors for predictions of milk protein by Equation [5.6]. Data in the top panel were uncorrected for the random effects of study, and data in the bottom panel were corrected.

experiments indicates the model correctly captured the response to each of the EAA.

Lean et al. (2018) also found that the NEAA were a significant determinant of milk protein production having a slope of 0.23 g milk protein/g metabolizable NEAA, which is almost double the estimate herein. A portion of the difference is explained by the method of estimating NEAA used by Lean et al. (2018). The assumed equivalence of AA mass in the free and protein bound forms,

when in fact, each peptide bond is formed by exclusion of a molecule of water results in approximately a 15% loss of mass when bound in protein form. Estimating NEAA as $\text{MPIn} - \text{EAA}$ as in Lean et al. (2018) results in approximately a 30% underestimate of the NEAA supply which would propagate to the slope estimate. All AA mass interconversions between peptide bound and free forms were appropriately adjusted herein. Such an adjustment of the Lean et al. (2018) estimate yields a slope

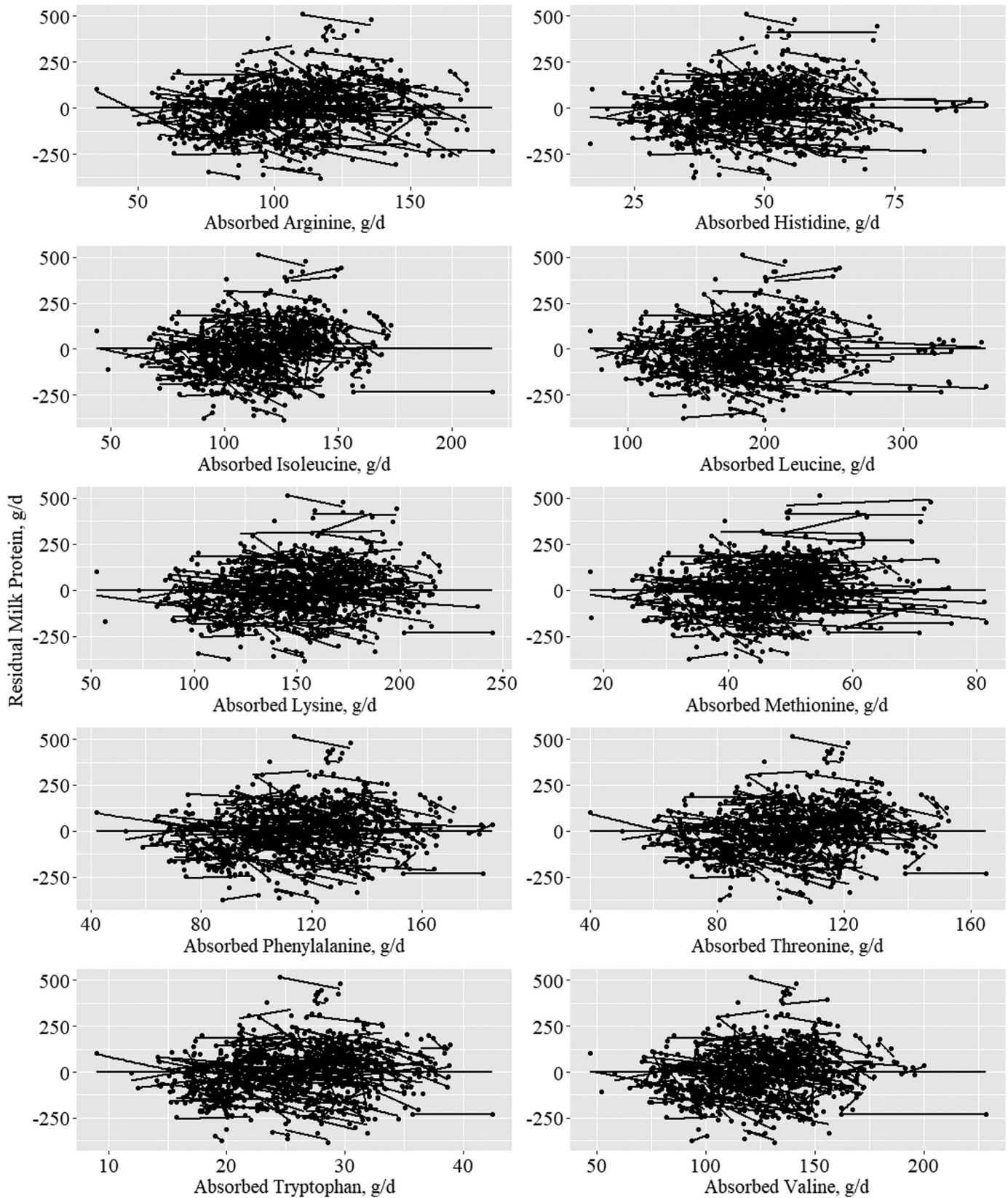


Figure 4. Milk protein residual errors for Equation [5.6] versus predicted absorbed supplies of EAA. Predictions and residuals were corrected for the random effect of study.

of 0.161 g/g, which is not different than the 0.114 g/g estimate for Equation [5.4].

Other experimental work (Metcalf et al., 1996; Doepel and Lapierre, 2010) does not support the observation of NEAA effects. Doepel and Lapierre (2010) infused 356 g NEAA/d into the abomasum with and without infused EAA and Metcalf et al. (1996) infused 192 g of NEAA into the jugular vein. Neither reported significant changes in milk protein output although Doepel and Lapierre (2010) did observe a numerical effect when the infusion was combined with additional EAA. The expected change in milk protein output based on Equation [5.4] was 40 g/d for Doepel and Lapierre (2010) and 22 g/d for Metcalf et al. (1996). Given an SEM of 37 g/d for the first and a standard error of difference of 14 g/d for the second, the expected response may not have been detectable.

Leucine is often considered the most potent stimulator of mTOR activity, and thus an important driver of protein synthesis (Lynch, 2001; Escobar et al., 2005). Rank order 172 (Equation [5.5]) includes an absorbed Leu term, although the slope estimate is relatively low at 0.37 g/g with a SE of 0.26 g/g. The relative error is large, but it should be noted that the SE is half of that for absorbed Ile and a third of that for His and Thr, and removal of Leu impacts the estimates of His, Ile and NEAA. Contrary to the conclusion reached by Lean et al. (2018) that Leu was a potent driver of milk protein synthesis, it seems more likely that Ile is exerting more influence on milk protein production than Leu under normal feeding conditions.

Differential responses to systemically infused EAA as compared with those absorbed from the intestine are likely real, but the magnitude is small enough that it is difficult to discern the effect. The $\text{InfTP}_{\text{Art}}$ term was only present in half of the top 5,000 models fitted to all of the data, and the slope estimate averaged 0.173 g of milk protein/g of infused EAA indicating a 17.3% first-pass loss of the EAA during absorption. The SE was as large as the estimate indicating the estimate does not significantly differ from 0. From prior work (MacRae et al., 1997; Hanigan et al., 2004; Fleming et al., 2019a), the utilization of EAA on first pass varies among EAA, but generally being 15% or less which is consistent with the estimate herein. Given the variance, much more data would be required to estimate the loss with any degree of precision using milk protein production as a response variable. Given that the presence or absence of the term in the model did not affect the remaining parameter estimates (Equation [5.5] vs. Equation [5.6]), it can be excluded from the model for field application.

Equation [5.7] was rank order 277 and represents the first model that did not include a term for Thr. As compared with Equation [5.5], its removal did not significantly alter the slope coefficients for Lys and Met, but it did increase the coefficients for His and Ile without

greatly altering their SE. The removal also resulted in a 1 unit increase in AIC, and a 1 g/d increase in RMSE. The changes in the His and Ile coefficients may indicate a lack of orthogonality, but the correlations among the slopes for each versus Thr are low suggesting that is not the case. It is unclear which estimates for His and Ile are more representative, but given the low correlations among the EAA parameter estimates, it is likely to be those from the more complicated model.

Intercept estimates ranged from -166 to 257 g/d across the 5,000 top models with a mean estimate of 44 g/d (Supplemental Table S4). However, screening and selection based on fit statistics and biology resulted in models that generally had slightly positive intercepts (Table 5) as compared with an expected value in the -150 to -200 g/d range as discussed above. This may reflect differences among the AA composition of milk protein (the response element defining both the intercept and the slope) and the composition of proteins representing the nonproductive functions. However, if this were the case, one would expect the global model, which contained all of the AA represented in MP, to have a similar intercept as for Equation [3.5]. It is also possible that the common quadratic term failed to fully capture the curvature in several of the EAA, but the intercept for Equation [4.1] was only slightly negative with most of the individual AA quadratics represented. Given that there were multiple models with the expected intercept among the population of models (Supplemental Table S4), it seems the problem is related to effects captured in AA that were screened which yielded slopes for the remaining AA that deviate slightly from the expected intercept. Given the apparent discontinuity between the intercepts for equations in Table 5 and factorial estimates (NASEM, 2021), predictions of milk protein below the range of data used herein to derive the equations should be evaluated for validity before use.

Cross-Evaluations. Equations [5.3] through [5.7] were subjected to cross-evaluation. Results from Equation [5.6] are presented in Table 6. Stability of parameter estimates generally aligned with the SE of the estimates in the original solves with the Leu, Thr, NEAA, and $\text{InfTP}_{\text{Art}}$ terms exhibiting large variation in estimates (CV >50%) across the solves, and mean P-values indicating they were not significantly different from 0. Terms for absorbed His and Ile had moderate CV, but were generally significantly different from 0, and the remainder excepting the intercept were generally stable across solves. Despite moderate variance in the estimates of model parameters, the resulting model evaluations were quite stable. Based on the marginal and conditional R^2 , approximately two-thirds of the remaining variance was associated with random study effects which would generally represent variation among studies in animal genetics, facilities and

Table 6. Results from cross evaluation of Equation [5.6] using 85% of the data for training and 15% for independent evaluations; data for each were randomly selected from the full dataset, and selection and evaluations were repeated 500 times

Variable ¹	Mean estimate	SD of mean	Mean SE	SD of SE	Mean CV	SD of CV	Mean <i>P</i> -value	SD of <i>P</i> -value
Intercept	32	52	103	2	874	4,230	0.64	0.21
BW – 612	–0.340	0.041	0.140	0.002	42	5	0.02	0.02
His	1.98	0.53	0.83	0.05	48	39	0.05	0.10
Ile	1.56	0.27	0.58	0.04	38	7	0.01	0.02
Leu	0.432	0.185	0.297	0.009	101	161	0.21	0.20
Lys	1.20	0.17	0.32	0.01	28	5	0.002	0.006
Met	1.71	0.20	0.43	0.02	25	3	0.0002	0.0005
Thr	1.62	0.62	0.91	0.09	72	112	0.13	0.15
NEAA	0.13	0.04	0.06	0.00	55	74	0.07	0.14
InfTP _{Art}	0.382	0.376	0.233	0.118	76	24	0.19	0.12
Σ(EAA ²)	–0.00218	0.00036	0.00049	0.00002	23	4	0.0002	0.0010
DEFAIn	10.36	0.79	1.92	0.05	19	1	0.000001	0.000003
DENDFIn	5.05	0.69	1.30	0.03	26	4	0.001	0.004
DErOMIn	11.85	0.59	1.52	0.05	13	1	0.000000	0.000000
DEStIn	10.68	0.46	0.90	0.02	8	0.4	0.000000	0.000000
Observed mean, g/d	923	17						
Predicted mean, g/d	927	12						
CCC	0.75	0.03						
RMSE, g/d	133	8						
RMSE, % mean	14.4	0.9						
Mean bias, %	0.7	0.9						
Slope bias, %	5.1	3.3						
Mean bias, g/d	–4	10						
Slope bias, g/g	0.192	0.077						
<i>P</i> _{Mean bias}	0.50	0.29						
<i>P</i> _{Slope bias}	0.07	0.15						

¹DEFAIn = digested energy intake from fatty acids (Mcal/d); DENDFIn = digested energy intake from NDF (Mcal/d); DErOMIn = digested energy intake from residual OM (Mcal/d); DEStIn = digested energy intake from starch (Mcal/d); InfTP_{Art} = the effect of systemic infusion of AA versus intestinal absorption; mean and slope bias % expressed as a % of the MSE; RMSE = root mean squared error (g/d or % of Obs mean).

facility management, environmental and disease stress, nutritional factors not captured as inputs to the model, and random errors of measurement (see Chapter 20 in NASEM, 2021). Thus, some of this variance could be explained through collection of a wider array of data, but some would remain as truly random, and not biologically explainable.

One challenge with several EAA was that relatively few studies have been conducted with dietary AA inputs independently manipulated by infusion or with a well-defined rumen-protected AA source. Therefore, the independent variation in inputs of those EAA was not well represented in the data. In selecting 85% of the data for training, several key treatments defining one of those EAA likely would have been excluded from the training set in one or more iterations resulting in poor estimates of those effects. Thus, additional experimental work is needed to gain precision and better evaluate the independent effects of several AA including Arg, Leu, Phe, Thr, Trp, and Val so that confidence in those effect estimates can be improved.

Exploration of Residual Errors. Residual errors were examined to determine whether any additional factors should have been considered in the models. From Figure 3, the residuals offer no evidence of slope bias and are

centered around 0. There also are no apparent patterns to the residuals within experiment although there are differences among experiments in the intercept which were adequately handled by the random effect of study as shown in the lower panel of Figure 3. There were also no clear patterns to the residuals when plotted against total EAA divided by DEInp indicating that the independent effects of these 2 terms were properly captured in the model. There were no obvious patterns to the residuals across or within studies when plotted against dietary concentrations of dFA, dNDF, dSt, or rOM or against BW. There was slight evidence for a potential negative slope to the residuals within study relative to increasing milk fat concentration, perhaps indicating that increasing energy use for milk fat reduced the yield of milk protein per unit of input (data not shown).

Consolidation of Equation [5.6]. Because the coefficients for DEStIn, DErOMIn, and DEFAIn were similar, and an infusion term was not needed for field use, a simplification of Equation [5.6] was undertaken. In addition to aggregation of the DE terms, the addition of the unused EAA to the NEAA term was explored, but contrary to the earlier NASEM (2021) work, we found that it provided no additional statistical benefit (data not shown), and thus, we did not include another AA term in

Table 7. Evaluation of the consolidation of terms in Equation [5.6] and a comparison to the equation used by NASEM, 2021

Term ²	NASEM (2021) ¹		[7.1]			[7.2]		
			NASEM (2021) refit			Consolidated Equation [5.6]		
	Estimate	SE	Estimate	SE	P <	Estimate	SE	P <
Intercept	−97	56	−31	57	0.59	−73	60	0.23
(BW-612)	−0.42	0.14	−0.36	0.14	0.01	−0.36	0.14	0.008
His, g/d	1.68	0.72	1.50	0.71	0.04	1.18	0.75	0.12
Ile	0.89	0.47	1.33	0.46	0.004	1.08	0.51	0.04
Leu	0.47	0.33	0.17	0.35	0.64	0.24	0.33	0.48
Lys	1.15	0.31	1.10	0.31	<0.001	1.08	0.32	0.001
Met	1.84	0.41	1.91	0.41	<0.001	1.91	0.41	<0.001
Thr						1.38	0.82	0.09
Other AA ³	0.077	0.038	0.071	0.043	0.10			
NEAA						0.074	0.051	0.15
Σ(EAA) ^{2,4}	−0.00215	0.00066	−0.00162	0.00070	0.02	−0.00175	0.00066	0.008
DEInp	10.8	0.70	9.63	0.73	<0.001			
DigNDF ⁵	−4.60	0.93	−5.29	0.93	<0.001			
DEFAIn + DErOMIn + DEStIn, mCal/d						10.8	0.73	<0.001
DENDFIn, mCal/d						5.41	1.2	<0.001
AIC	N/A ^c		10,399			10,405		
σ ²			6,750			6,800		
τ ₀₀ (PubID)			16,663			16,533		
ICC			0.71			0.71		
Marginal R ²			0.49			0.49		
Conditional R ²			0.85			0.85		
Predict mean, g/d	915		926			926		
CCC	0.75		0.75			0.75		
RMSE, g/d	134		133			133		
RMSE, % of mean	14.6		14.5			14.4		
Mean bias, % of MSE	0.3		0.1			0.1		
Slope bias, % of MSE	3.4		4.0			4.1		
Mean bias, g/d	7		−4			−4		
Slope bias, g/g	0.165		0.178			0.181		
P _{Mean bias}	0.11		0.4			0.4		
P _{Slope bias}	0.0001		0.0001			0.0001		

¹NASEM 2021 model was derived using a prior version of the dataset before additional cleaning, thus the parameter estimates reflect the earlier data, and the AIC and random effect statistics are not comparable to those derived from the current data. The RMSE analyses reported here reflect performance of that equation using the revised dataset herein.

²DEInp = digested energy intake minus the DE associated with absorbed protein (Mcal/d); DEFAIn = digested energy intake from fatty acids (Mcal/d); DENDFIn = digested energy intake from NDF (Mcal/d); DErOMIn = digested energy intake from residual OM (Mcal/d); DEStIn = digested energy intake from starch (Mcal/d); DigNDF = digested NDF (% of DM); dSt = digested starch (% of DM); RMSE = root mean squared error (g/d or % of Obs mean).

³Summation of other EAA not explicitly represented plus the NEAA.

⁴Summation of the squared individual, absorbed EAA explicitly represented in each equation, e.g., His, Ile, Leu, Lys, Met, and Thr (g²/d).

⁵Derived before additional data cleaning and not re-calculated.

place of the NEAA term. The simplified model, Equation [7.2] is provided in Table 7 along with an evaluation of the published NASEM (2021) equation and the NASEM (2021) equation refitted to the data. The NASEM (2021) equation was derived earlier in the analytical process, and thus represents our exploration of the potential differential effects of energy coming from differing substrates using nutrient concentrations in the diet. We later discovered this could be more cleanly addressed using the DE terms for each which also negated the remaining collinearity problems for the DEInp term. The lesser slope for DENDFIn than for other DE terms is more easily interpreted in this form than when using an aggregated DEInp term and adjustment slopes based on dietary NDF

concentrations. The NASEM (2021) equation was also derived before more data screening uncovered additional entry errors that were corrected, hence the refit of the prior model form to the current data to allow like comparisons. Based on RMSE, the refitted NASEM (2021) equation performed the same as the original equation; however, the slope estimates for absorbed His and Leu declined and the estimate for absorbed Ile increased.

Aggregation of terms yielding Equation [7.2] increased the AIC very slightly to 10,405, but the RMSE only increased 1 g/d. Thus the consolidated model performed as well as Equation [5.6], and as well as the equation provided in NASEM (2021). Not only do these models perform equally well, but there are many additional

equations that perform similarly. Some were ruled out for biological reasons, but many additional ones are potentially valid.

Given the superiority of the equations based on independent and additive responses to the individual nutrients as compared with those based on first limiting nutrient principles (Table 4) and the extensive mechanistic work (Cant et al., 2003; Anthony et al., 2004; Burgos and Cant, 2010; Appuhamy et al., 2011b, 2012, 2014a; Arriola Apelo et al., 2014c,d; Doelman et al., 2015a,b; Liu et al., 2017; Nichols et al., 2017; Yoder et al., 2020), the first limiting nutrient hypothesis must be dismissed. The disadvantage of the revised representation of nutrient effects is that it is not possible to define requirements for any of the terms in a classical sense. Previous models (NRC, 1966, 1971, 1978, 1989, 2001) were based on assumptions of fixed efficiencies of nutrient use with no additivity of responses, which is clearly not consistent with the observed data (Supplemental Figure S4, see Notes). Under those assumptions, unique requirements for each nutrient could be algebraically determined. Given observed independent, additive, and approximate linear responses throughout the normal feeding range, there are an infinite number of solutions that provide the same level of production, and the best diet will be the one that provides an optimum mix of nutrients that achieves the desired level of performance at a minimum cost.

Although much of the variance is captured in these equations, the random effect of study still accounts for approximately 35% of the overall variation (Table 5). Some of this variation is truly random errors of measurement, but a portion will be due to environmental effects, health challenges, and genetic influences (NASEM, 2021). As these cannot currently be captured in the predictions *a priori*, the model may exhibit mean bias when applied to individual farms and pens within farms, but the predictions should be representative across a range of farms and pens. The plateau defined by the quadratic term in the equations is also problematic for farm application as it represents an apparent maximum milk protein production for animals used in studies over the past 30 yr, which is generally well below that achieved on well managed farms today. As genetics and management at a particular farm change slowly over time, provision should be made for intercept and plateau adjustments by farm as outlined by NASEM (2021) to accommodate farm specific effects until predictions of those effects can be included in the system.

CONCLUSIONS

The existing data were adequate to define the effects of DE, Ile, Lys, and Met on milk protein production

with good precision, the effects of His and the NEAA with moderate precision; and the effects of Leu and Thr with poorer precision. Milk protein production equations based on additive and independent effects of the individual AA and energy providing nutrients performed much better than equations based on 1st limiting nutrient concepts. Milk protein responses to supplies of MP and total EAA were curvilinear yielding variable efficiencies of conversion of absorbed EAA to milk EAA. Linear responses to individual EAA were observed, but quadratic responses were not well defined by the data. Expression of EAA as percentages of MP or as a ratio to DEIn resulted in slightly poorer model performance, and given such ratios also violate linearity assumptions relied upon to derive the regressions, those forms cannot be recommended.

NOTES

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Nonstandard abbreviations used: σ^2 = mean random effect variance; τ_{00} = between group variance; AIC = Akaike information criterion; CCC = concordance correlation coefficient; CPIn = CP intake; DE = digested energy; DEFAIn = DEIn was parsed to the individual contributions from fatty acids; DEIn = digested energy intake; DEInp = nonprotein digested energy intake; DENDFIIn = DEIn from NDF; DEROMIn = DEIn from residual OM; DESTIn = DEIn from starch; DETPIIn = DEIn from true protein; EAAIn = EAA intake; FA = fatty acids; ICC = interclass correlation coefficient; MPIn = MP intake; MSE = mean squared error; NA = not available; NP = net protein; Obs = observed; Pred = predicted; PubID = publication identification; RDNDF = ruminally digested fiber; RDSt = ruminally digested starch; RMSE = root mean squared error; rOM = residual OM; TP = true protein.

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